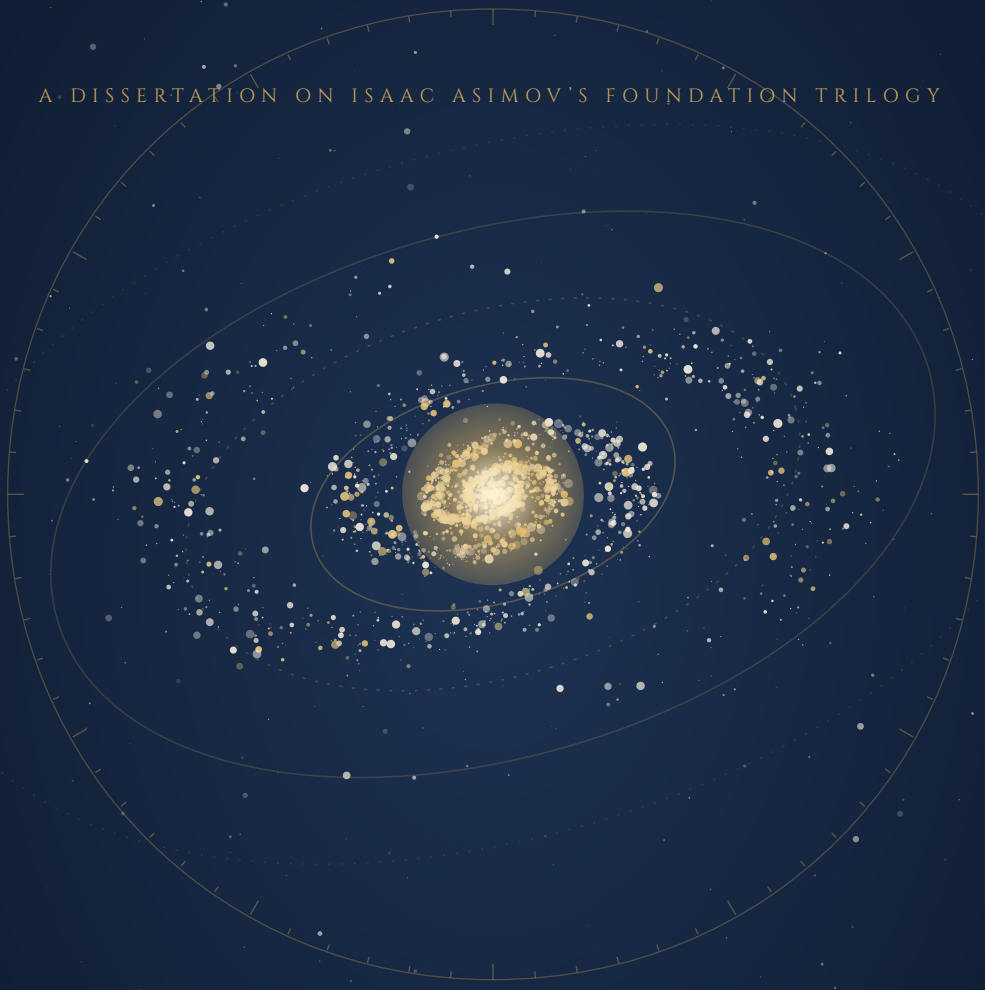
A DISSERTATION ON ISAAC ASIMOV'S FOUNDATION TRILOGY



THE AUTOMATON AND THE DWARF

Psychohistory, Political Theology, and the Invention of Stochastic Sovereignty in Isaac Asimov's Foundation Trilogy

OCTOBER 2026

FRONT MATTER

Abstract

This dissertation advances a new interpretation of Isaac Asimov's Foundation Trilogy (*Foundation*, 1951; *Foundation and Empire*, 1952; *Second Foundation*, 1953), assembled from nine narratives serialized in *Astounding Science Fiction* between 1942 and 1950, as the founding scripture of what it terms **stochastic sovereignty**: a form of rule exercised not over bodies, territories, or legal acts but over the probability distributions of populations, and whose constitutive condition is a managed asymmetry of knowledge between the governing and the governed. Against the two dominant critical traditions, which read psychohistory either as a scientistic fantasy of prediction or as historical materialism distorted into cyclical elitism (Elkins 1976), the study argues that the Trilogy executes a strict dialectic among three epistemic demons of modernity: Laplace's demon of total prediction, which presides over the self-executing Plan of the first volume; Maxwell's demon of local, information-dependent sorting against entropy, which the Second Foundation becomes once the Plan must be maintained against perturbation; and Descartes's *malin génie*, whose systematic manufacture of false belief completes the Second Foundation's final stratagem. Their synthesis is named here the **Seldonian Demon**. Taking as its governing emblem Walter Benjamin's 1940 allegory of the chess-playing automaton secretly operated by a hidden dwarf, the dissertation reconstructs the genealogy of each component, from Ibn Khaldun, Vico, Montesquieu, Gibbon, Condorcet, Quetelet, Maxwell, Tolstoy, Marx, Plekhanov, Spengler, and Toynbee to the political theology of Augustine, Joachim of Fiore, the Leibniz–Clarke correspondence, Schmitt, Löwith, Blumenberg, Voegelin, and Agamben, while distinguishing rigorously among attested influence, contextual synchrony, and structural homology.

Five findings follow. First, psychohistory's second axiom, which requires the predicted population to remain ignorant of the prediction, is shown by way of the fixed-point theorems of Simon (1954) and Grunberg and Modigliani (1954) to be mathematically dispensable, and therefore to be a political *arcanum imperii* presented as a methodological constraint: secrecy is the condition not of predictability but of discretion. Second, the Mule is reread as the *amplified singular*, a quantum-genetic contingency in Schrödinger's and Jordan's sense that occupies the structural position of Schmitt's sovereign exception and of Dostoevsky's Underground Man, and whose defeat obliges the Second Foundation to become a distributed Mule. Third, the Second Foundation reconstitutes, in the idiom of cybernetics, the providential machine of

Christian theology, executing a general providence (the Plan) through a special providence administered by its Speakers. Fourth, every reduction of historical entropy in the Trilogy incurs what the study calls a *Landauer cost*, paid in erased persons, rewritten minds, and falsified collective memory, a cost that can be displaced but never abolished. Fifth, a comparative analysis spanning Bacon, Borges, Hesse, Stapledon, Heinlein, Miller, Herbert, the Strugatskys, Lem, Le Guin, Dick, Banks, and Liu Cixin yields a taxonomy of four literary responses to the Seldonian Demon: providential, apophatic, insurgent, and tragic-progressor.

The dissertation concludes that psychohistory is a promise that has forgotten it is a promise. What sustains the Plan inside the fiction is not mathematical truth but intergenerational fidelity, renewed in the liturgy of the Time Vault. Drawing on Arendt's account of promising as the remedy for the unpredictability of action, and on Kant's publicity test of right, it proposes **covenantal contingency** as the normative alternative to stochastic sovereignty, an alternative made urgent by an age of population-scale predictive systems, algorithmic governance, and longtermist philanthropy that have openly adopted the Trilogy as their charter myth.

FRONT MATTER

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CHAPTER

I

Introduction

The Interregnum as Problem

> evulgato imperii arcano, posse principem alibi quam Romae fieri. *The secret of empire had been divulged: that an emperor could be made elsewhere than at Rome.*
>
> Tacitus, *Histories* 1.4

1.1 The Object

The Foundation Trilogy occupies an anomalous position in the history of the novel of ideas. It is at once the most influential work of speculative fiction ever devoted to the philosophy of history and a text whose formal properties seem designed to frustrate the conventions by which novels of ideas are read. Its nine constituent narratives appeared in John W. Campbell's *Astounding Science Fiction* between May 1942 and January 1950; Gnome Press gathered them into three volumes between 1951 and 1953, adding a newly written opening, "The Psychohistorians," to frame the whole. In 1966 the World Science Fiction Convention awarded the sequence a one-time Hugo for Best All-Time Series, ahead of Tolkien's *The Lord of the Rings*, a verdict that registers less the work's literary finish than the degree to which it had become the genre's default model of galactic history. Its protagonists are replaced every generation; its climaxes are conversations rather than battles; its prose, which Asimov likened to plate glass rather than stained glass, renounces ornament for argument. The unit of narration is neither the individual life nor the campaign but the *crisis*, a structural constriction of historical possibility, and the true protagonist is a mathematical object that never appears on the page.

The premise can be stated briefly. In the twelfth millennium of a Galactic Empire of some twenty-five million inhabited worlds, the mathematician Hari Seldon employs psychohistory, a statistical science of mass behaviour, to predict the Empire's collapse and an ensuing interregnum of thirty thousand years. Tried before the Commission of Public Safety, he converts his sentence of exile into an instrument of policy. Two Foundations are established at "opposite

ends" of the Galaxy: the first, ostensibly charged with compiling an *Encyclopedia Galactica*, on the metal-poor peripheral world of Terminus; the second at a location withheld from the reader. By shepherding the first Foundation through a sequence of "Seldon Crises," the Plan will shorten the interregnum to a single millennium. The Foundation passes through successive hegemonies, scientific, sacerdotal, commercial, and plutocratic, until a mutant conqueror, the Mule, whose power over human emotion lies outside psychohistory's computational horizon, overturns the Plan. The Plan is restored only through the covert agency of the Second Foundation, a society of "mentalic" psychologists who, the last page discloses, have resided all along on Trantor, the ruined imperial capital itself.

1.2 The State of the Question

Critical writing on the Trilogy divides into three traditions, each of which captures a stratum of the text and none of which explains its internal reversal. The *scientific-humanist* tradition, represented by Joseph Patrouch's *The Science Fiction of Isaac Asimov* (1974) and James Gunn's *Isaac Asimov: The Foundations of Science Fiction* (1982), reads psychohistory as an extrapolation of the social sciences and the series as a rationalist celebration of intelligence over violence. The *ideology-critical* tradition finds its canonical statement in Charles Elkins's "Isaac Asimov's *Foundation* Novels: Historical Materialism Distorted into Cyclical Psycho-History" (*Science Fiction Studies*, 1976), which argues that Asimov evacuates historical materialism of class struggle and praxis, substituting an elite of manipulators and a fatalistic cycle that mirrors the American middle class's experience of history as something that happens to it. Darko Suvin's account of space opera as feudal costume drama transposed to the stars (*Metamorphoses of Science Fiction*, 1979) belongs in spirit to the same tradition. The *allegorical* tradition reads the Trilogy as Gibbon in space: Trantor as Rome, Bel Riose as Belisarius, the Periphery as the barbarian West. Donald Wollheim's *The Universe Makers* (1971) generalised the point, treating the Trilogy as the template of a consensus future history in which the rise, fall, and rebirth of a galactic empire became the genre's shared myth.

Each tradition is correct about what it sees. The scientific-humanist reading correctly identifies the work's ethical preference for cunning over force; the ideology-critical reading correctly identifies its elitism; the allegorical reading correctly identifies its sources. What none explains is the most consequential fact about the work's architecture: that a narrative which begins in 1942 as a fable of prediction ends in 1950 as a fable of administration. In the first volume the Plan executes itself, and the Foundation's leaders need only recognise the single path each crisis leaves open. By the third, the Plan has become a fragile artefact maintained by a hidden priesthood that edits minds, sacrifices its own members, and deceives the very civilisation it serves. The book that opens under the sign of Laplace closes under the sign of Dostoevsky's Grand In-

quisitor. This dissertation takes that reversal as its central object and argues that it is neither an accident of serial composition nor a failure of nerve, but the rigorous unfolding of the premise.

1.3 Thesis

The Foundation Trilogy is the founding scripture of *stochastic sovereignty*. By this term I designate a form of rule with four distinguishing marks. Its object is not the individual body, as in classical sovereignty; nor the norm, as in juridical power; nor even the population as a biological aggregate, as in Foucault's biopolitics; but the *probability distribution* of a population's possible futures. Its instrument is not command but the modulation of initial conditions, information states, and affective dispositions. Its legitimacy derives from a claim to have computed in advance the trajectory that minimises aggregate suffering. Its constitutive condition, the condition without which, the Trilogy insists, it cannot function at all, is the ignorance of the governed concerning the fact, the content, and the agents of their governance.

The Trilogy reaches this form of rule through a dialectic among three demons that modern thought invented to dramatise the limits of knowledge. *Laplace's demon*, the intellect that would comprehend all forces and positions and hence the entire future, presides over the first volume, in which Seldon's recorded image appears in the Time Vault to announce what has already been computed. *Maxwell's demon*, the being who sorts swift molecules from slow and thereby reverses entropy locally at the price of information, presides over the Second Foundation once the Plan must be actively maintained against perturbation. *Descartes's evil genius*, the deceiver who fabricates a subject's entire world of belief, presides over the denouement, in which the Second Foundation persuades the First that it no longer exists. The synthesis of the three, an agency that predicts, intervenes, and deceives, whose predictions depend on its interventions and whose interventions depend on its deceptions, I call the *Seldonian Demon*.

Four subsidiary theses organise the argument. The *arcanum thesis* (Chapter VIII) holds that psychohistory's axiom of ignorance is not, as the text presents it, a methodological necessity but a political arcanum: the fixed-point results published in 1954 by Herbert Simon and by Emile Grunberg and Franco Modigliani establish that self-confirming public predictions exist under broad conditions, so that secrecy is chosen because it enlarges the controller's discretion, not because prediction demands it. The *exception thesis* (Chapter VI) holds that the Mule occupies precisely the structural position of Schmitt's sovereign exception and of the theological miracle, so that the Second Foundation, in defeating him, must become a distributed Mule. The *providence thesis* (Chapter VII) holds that the Second Foundation reconstitutes, in the vocabulary of cybernetics, the providential machine of Christian theology, replaying the Leibniz–Clarke controversy in reverse. The *Landauer thesis* (Chapter IX) holds that every reduction of historical entropy in the Trilogy carries an irreducible cost in erasure, borne by persons and

memories the narrative quietly writes off. Chapter XII converts these diagnoses into a normative proposal, *covenantal contingency*, derived from Arendt's analysis of promising.

1.4 The Emblem

The governing emblem of this study is the first of Walter Benjamin's theses "On the Concept of History," composed in 1940, two years before the first Foundation story. Benjamin recalls the celebrated chess automaton of the eighteenth century: a puppet in Turkish dress seated at a board, which wins every game, while a system of mirrors makes the table appear transparent and conceals a hunchbacked dwarf, a master of the game, who guides the puppet's hand from within. In Benjamin's allegory the puppet is historical materialism, and the dwarf is theology, which can win only if it stays out of sight. The Trilogy realises this allegory with uncanny literalness. Its automaton is the visible machinery of the Plan, most concretely the recorded Seldon who materialises at fixed intervals in the glass cubicle of the Time Vault, seated in his wheelchair, to pronounce on crises he cannot see. Its dwarf is the Second Foundation, a theology of mind concealed inside a materialist mechanism and directing its moves. That the Trilogy's other decisive figure, the Mule, enters the narrative disguised as a gangling, grotesque clown completes the emblem: the hidden master and the deformed outsider are two faces of a single power.

1.5 Method: Influence, Synchrony, Homology

Because this study brings Asimov into dialogue with thinkers he never read, it must state precisely what kind of relation each comparison asserts. Three registers are distinguished. *Attested influence* is confined to sources for which documentary evidence exists: Gibbon, whose *Decline and Fall* Asimov, by his own account, had read twice before conceiving the series on 1 August 1941; the editorial pressure of Campbell, who by Asimov's later testimony urged him to break the Plan, and whose conviction of human superiority over any imaginable alien led Asimov to populate his Galaxy with human beings alone; and the kinetic theory of gases, the statistical mechanics in which Asimov, who received his doctorate in chemistry from Columbia in 1948, was professionally trained. *Contextual synchrony* designates works produced in the same years under shared discursive conditions, without direct contact: Popper's "The Poverty of Historicism" in *Economica* (1944–45); Hayek's essays on "scientism" (1942–44); Schrödinger's *What Is Life?* (1944); Wiener's *Cybernetics* and Shannon's information theory (1948); Merton's "The Self-Fulfilling Prophecy" (1948); Braudel's *La Méditerranée* and Löwith's *Meaning in History* (1949); Heinlein's "Gulf," which shared two issues of *Astounding* with the final Foundation serial; and Hubbard's "Dianetics," which followed it into Campbell's pages in May 1950. *Structural homology* designates parallels of form with earlier or later works, from Ibn Khaldun

and Vico to Joachim of Fiore, Bacon, and the rabbinic academy at Yavneh, that illuminate the Trilogy's logic without implying descent. Homology is not a weaker species of influence but a different kind of claim: it asserts that the Trilogy reached, by its own narrative necessity, positions the tradition had reached before it, and that the recurrence is itself evidence about the structure of the problem.

The corpus is the Trilogy as published between 1951 and 1953. Asimov's later sequels and prequels (1982–1993), which recast the Plan as one stage in a design by the robot R. Daneel Olivaw and subordinate it to the collective mind of Gaia, are treated as apocrypha: evidence of the tensions the Trilogy left unresolved, not constraints on its interpretation. The method throughout is that of the *constellation* in Benjamin's and Adorno's sense: concepts drawn from heterogeneous traditions are arranged around the text until their configuration renders legible what none could disclose alone.

1.6 Plan of the Study

Chapter II reconstructs the archaeology of psychohistory, from providential statistics to the kinetic theory of gases and its return to society. Chapter III situates the Trilogy among the great morphologies of decline. Chapter IV analyses the instrumental use of the sacred and of commerce in the Foundation's ladder of hegemony. Chapter V examines the logic of the Seldon Crisis and the paradox of narrating determinism through clever protagonists. Chapter VI reads the Mule as the amplified singular. Chapter VII reads the Second Foundation as a providential machine. Chapter VIII dismantles the axiom of ignorance. Chapter IX develops the thermodynamic ethics of the Landauer cost. Chapter X constructs a comparative taxonomy of speculative and philosophical fiction. Chapter XI traces the afterlife of psychohistory in the technocratic imaginaries of the present. Chapter XII proposes covenantal contingency.

CHAPTER

II

The Differential of History

An Archaeology of Psychohistory

> *"...the differential of history, that is, the individual tendencies of men..."*
>
> Tolstoy, *War and Peace*, vol. III, pt. 3, ch. I, trans. Louise and Aylmer Maude

2.1 The Definition and Its Axioms

The Trilogy defines psychohistory not in Seldon's voice but in the impersonal voice of the *Encyclopedia Galactica*, whose entries serve as epigraphs to its chapters. In paraphrase, the entry holds that psychohistory is the branch of mathematics concerned with the reactions of human conglomerates to fixed social and economic stimuli, and that its every formulation presupposes two conditions: that the conglomerate be large enough for valid statistical treatment, and that it remain unaware of the analysis, so that its reactions are truly random. Three features of this definition have received less attention than they deserve.

The first is that the second axiom makes randomness a product of ignorance. The randomness on which psychohistory depends is not an ontological indeterminacy of human action but a statistical independence between the population's behaviour and the predictor's knowledge, and that independence is secured by withholding information. Ignorance, in short, manufactures the very randomness that the science then exploits. The second feature is a concealed third axiom: the constancy of human nature across the forecast horizon. This is the oldest premise of political historiography, from Thucydides's claim (1.22) that his work would be a possession for all time because human nature would produce like events again, to Hume's remark in the first *Enquiry* (§8) that "mankind are so much the same, in all times and places, that history informs us of nothing new or strange in this particular." Psychohistory inherits the axiom intact, and the Mule violates it, for he is precisely a change in human nature.

The third feature is closure. A statistical mechanics requires an isolated system, and psychohistory requires a Galaxy in which no non-human agency perturbs human aggregates. Asimov's decision to exclude extraterrestrials, which he attributed to a wish to avoid Campbell's

insistence that human beings must always prevail over alien species, thereby acquired an epistemological function its author did not intend: a contingent editorial accommodation became the closure condition of the model. Robots, too, are absent from the Trilogy. Their reinsertion in the late sequels as the concealed authors of the Plan is the apocrypha's most revealing confession: the closure was never real, and the system had been open all along to an agency outside the human conglomerate.

Nor was the closure ideologically neutral. Campbell's insistence on human supremacy over every imaginable alien belonged to an openly hierarchical outlook whose racial form he later put into print, in editorials and correspondence that his biographer Alec Nevala-Lee has documented (*Astounding*, 2018); in 2019 the award for new writers that had borne Campbell's name since 1973 was renamed after its recipient, Jeannette Ng, denounced his politics in her acceptance speech. Asimov's all-human Galaxy evaded that chauvinism rather than refuting it, and the evasion has a structural consequence. The human conglomerate on which psychohistory operates is constituted by an exclusion, and what the model excludes, non-human intelligence, returns in the apocrypha as its hidden sovereign.

2.2 Statistics as Theodicy

The genealogy of psychohistory is usually traced to positivist sociology. It begins earlier, and in theology. In 1710 John Arbuthnot presented to the Royal Society "An Argument for Divine Providence, taken from the constant Regularity observ'd in the Births of both Sexes." Examining London christenings over eighty-two consecutive years, he found male births exceeding female births in every one, computed the probability of such a run under pure chance as one in two raised to the eighty-second power, and concluded that the regularity was the work of a Providence supplying surplus males to offset their higher mortality. Historians of statistics regard the paper as the first test of significance. In 1741 Johann Peter Süssmilch generalised the argument in *Die göttliche Ordnung in den Veränderungen des menschlichen Geschlechts*, the founding text of German demography, whose title declares that the regularities of births, marriages, and deaths are the visible signature of a divine order.

The consequence for the Trilogy is considerable. Psychohistory does not secularise providence; it returns statistical reason to its theological birthplace. The first modern inference from aggregate regularity was an inference to a hidden governor. The Trilogy's final revelation, that the regularities of Galactic history are sustained by a concealed order of guardians, repeats Arbuthnot's inference and literalises it. The hidden governor exists, and he keeps an address on Trantor.

2.3 Social Mathematics under the Terror: Condorcet and Laplace

The programme of a predictive science of society received its first manifesto from a mathematician in hiding. Condorcet composed the *Esquisse d'un tableau historique des progrès de l'esprit humain* in 1793–94 while proscribed by the revolutionary government, and its tenth epoch argues that if the natural sciences can predict phenomena whose laws they know, nothing renders it chimerical to sketch, with some probability, the future destiny of the human species from the results of its history. He called the requisite discipline *mathématique sociale*. Proscribed by the National Convention in 1793 after attacking the Montagnard constitution, Condorcet was arrested at Clamart in late March 1794 and found dead two days later in his cell at Bourg-la-Reine; whether he took poison, died of exhaustion, or was killed has never been established. He died under the Terror over which the Committee of Public Safety then presided. The body that tries Hari Seldon is the Commission of Public Safety. Whether or not Asimov intended the echo, the scene restages the predicament of Enlightenment social mathematics under emergency rule and reverses its outcome. Condorcet wrote the future in hiding and perished; Seldon computes the reactions of his judges, negotiates the terms of his own exile, and makes the Commission the unwitting founder of his project. The dangerous mathematician survives by predicting the tribunal.

Condorcet's fate has a twentieth-century counterpart. Nikolai Kondratiev, whose 1925 theory of long waves modelled capitalist economies as cycles of roughly half a century, was arrested by the Soviet security police in 1930 and shot in September 1938, during Stalin's Great Terror. Condorcet and Kondratiev are the failed Seldons of modern history: mathematicians of social time destroyed by the states whose futures they computed. Seldon is the one who survives, because he computes the state itself.

Laplace supplied the programme's metaphysics and its method. The *Essai philosophique sur les probabilités* (1814) imagines an intelligence that, knowing at a given instant all forces and all positions, would embrace in a single formula the motions of the greatest bodies and the lightest atoms, so that the future, like the past, would lie open to its gaze. Yet the same treatise insists that probability is relative partly to our ignorance and partly to our knowledge, and applies the calculus to testimony, tribunals, and elections. Psychohistory unites these two Laplaces, the ideal of total determination and the practice of probabilistic governance under ignorance. The Seldon of the first volume is the demon; the Speakers of the third are the probabilists.

2.4 The Chiasmus of Statistical Reason

Adolphe Quetelet's *Sur l'homme et le développement de ses facultés, ou Essai de physique sociale* (1835) constructed the *homme moyen* and observed that crime is paid each year like a budget, with terrifying regularity. Its most notorious sentence, in my translation, holds that it is society

that prepares the crime, and the guilty party is merely the instrument that executes it. Comte, finding that Quetelet had appropriated the phrase "social physics," coined *sociologie* for his own enterprise. Henry Thomas Buckle's *History of Civilization in England* (1857–61) drew from the constancy of suicides, marriages, and even misdirected letters the conclusion that history obeys laws before which individual volition is negligible.

Then the method crossed disciplines. James Clerk Maxwell read John Herschel's 1850 *Edinburgh Review* essay on Quetelet and, in 1859–60, modelled the molecules of a gas as a population with a distribution of velocities. Theodore Porter and Ian Hacking have documented how the statistical method entered physics from social science rather than the reverse, and Boltzmann repeatedly justified his reasoning by analogy with the statistics of human populations. Maxwell himself, in an 1873 paper on determinism and free will read to the Eranus Club at Cambridge, located freedom at *singular points*: configurations at which an infinitesimal variation determines a finite divergence, as when a boulder poised on a ridge may fall to either side, and which multiply as systems grow more complex. Chapter VI will argue that the Mule is a Maxwellian singular point of Galactic history.

Asimov, trained in physical chemistry and the author of a doctoral dissertation on the kinetics of an enzyme reaction, sent the metaphor home. Psychohistory carries kinetic theory back to the society from which Maxwell had borrowed it. I call this double passage, from society to gas and from gas to society, the *chiasmus of statistical reason*, and its significance lies in what changed at each crossing. For Quetelet regularity was moral: the average man was the nation's ideal type, and deviation was error. For Maxwell it was epistemic: macroscopic law was microscopic ignorance averaged into certainty. For Asimov it becomes political: regularity is something that can be produced, preserved, and owned. The Second Foundation does not merely observe the law of large numbers; it administers it.

2.5 Tolstoy's Integral

The closest anticipation of psychohistory is not sociological but novelistic. In *War and Peace* (1869) Tolstoy argues that history can discover its laws only by abandoning the wills of heroes for the infinitesimal tendencies of ordinary persons and learning the art of integrating them; that the king is the slave of history; and that Napoleon's apparent genius is an optical effect of the masses' movement, while Kutuzov's wisdom consists in patient non-interference with forces he cannot command. Psychohistory is Tolstoy's integral actually computed. Isaiah Berlin's "The Hedgehog and the Fox" (1953) diagnosed in Tolstoy a fox by temperament who believed in being a hedgehog, a perceiver of irreducible particulars who longed for a single explanatory principle. The Trilogy has the same divided constitution in inverted form: a hedgehog's thesis, one Plan governing all, delivered in a fox's form, a dozen episodes and as many protagonists. Chapter V will treat this division as the work's generative contradiction.

2.6 The Absent Equation

The Trilogy never shows its mathematics. In *Second Foundation* the Prime Radiant projects the Plan as luminous equations across a chamber's walls, with deviations marked for correction, yet not one symbol is transcribed. In Istvan Csicsery-Ronay's terms, psychohistory is a fictive neologism standing in for an imaginary science: a name occupying the place of a theory. The absence is not a defect of imagination but the formal correlate of the second axiom. The reader belongs to the population, and the population must not see the equations. The Prime Radiant is, moreover, an image of what Joseph Frank, writing in 1945, called spatial form: the whole of a temporal sequence held present at once. A first reading places one in the position of the Foundation, surprised by each crisis; a rereading places one in the position of the Speakers, for whom the sequence is a single figure. The Trilogy thus teaches, in its form, the difference between living inside the Plan and surveying it.

2.7 The Contemporaneous Refutation: Popper and Hayek

While "Dead Hand" and "The Mule" were appearing in *Astounding*, Karl Popper was publishing "The Poverty of Historicism" in *Economica* (1944–45), later issued as a book (1957) dedicated to the victims of belief in inexorable laws of historical destiny. Its mature argument runs: the course of history is strongly influenced by the growth of knowledge; the future growth of knowledge cannot be predicted by rational means; therefore the course of history cannot be predicted. Popper distinguished unconditional prophecy from conditional scientific prediction and named the influence of a prediction on the predicted event the *Oedipus effect*. Psychohistory is very nearly the doctrine Popper refuted. The Trilogy, however, anticipates the refutation twice over. It constrains the growth of knowledge: the Foundation's science is curated, its priests maintain reactors they do not understand, and its technology is channelled by scarcity into miniaturisation. And it abandons unconditional prophecy for continuous correction. The result is a hybrid Popper did not foresee: holistic ends pursued by piecemeal means, utopian engineering executed through the incremental adjustments he commended to liberal reform, but conducted in secret.

Friedrich Hayek's essays on "scientism" (*Economica*, 1942–44) and "The Counter-Revolution of Science" (1941) indicted the engineering ambitions of Saint-Simon and Comte, and "The Use of Knowledge in Society" (1945) argued that the knowledge of particular circumstances of time and place is dispersed among millions and cannot be centralised, so that only the price system can coordinate it. This was Hayek's contribution to the socialist calculation debate opened by Mises in 1920 and answered by Oskar Lange's model of market socialism (1936–37), in which a Central Planning Board adjusts parameters by trial and error in response to market signals. Psychohistory is an imaginary resolution of that debate. At Galactic scale the

dispersed knowledge no planner could gather cancels into statistical regularity, so that the planner requires only the aggregate. The Trilogy's institutional architecture is accordingly Langean: the traders and merchant princes of the First Foundation are decentralised market agents whose initiative the Plan presupposes, while the Second Foundation is a concealed Central Planning Board that monitors aggregate outcomes and corrects parameters by trial and error. The Trilogy may be the only major fiction of the calculation debate, and it settles the debate by fiat of scale.

2.8 Coda: Cliodynamics as Experiment

Peter Turchin's cliodynamics (*Historical Dynamics*, 2003; with Sergey Nefedov, *Secular Cycles*, 2009; *Ages of Discord*, 2016), building on Jack Goldstone's structural-demographic theory (1991), models secular cycles of population pressure, elite overproduction, and state fiscal distress, and both its practitioners and its critics routinely invoke psychohistory as its fictional ancestor. One difference is decisive. Cliodynamics publishes its forecasts. Its existence is therefore a running experiment on the second axiom, to which Chapter VIII returns.

2.9 The Mathematical Sublime

The Trilogy's numbers, twenty-five million worlds, a quadrillion persons, thirty thousand years, are built to defeat the imagination. Kant's *Critique of the Power of Judgment* (1790) defines the mathematical sublime as the experience in which imagination fails to comprehend a magnitude in a single intuition while reason thinks it as a totality, so that the failure of the senses discloses the superiority of the rational faculty. Psychohistory is the mathematical sublime converted into a method of government. The Galaxy is too large to picture but not too large to compute, and the pleasure the Trilogy offers, which Csicsery-Ronay calls the science-fictional sublime, is the pleasure of reason triumphing over imagination's defeat. Its political lesson is double. The faculty that makes the totality thinkable makes the individual invisible, because in the sublime of number no single person can be seen. Chapter XII returns to this blindness as the difference between a *who* and a *how many*.

CHAPTER

III

Morphologies of the Imperial Corpse

Decline, Cycle, and the Interregnum

> *"...the decline of Rome was the natural and inevitable effect of immoderate greatness."*
>
> Gibbon, *Decline and Fall of the Roman Empire*, "General Observations on the Fall of the Roman Empire in the West"

3.1 Gibbon and the Long Fall

Gibbon is the one source of the Trilogy whose influence is beyond dispute. Asimov recounted that on 1 August 1941, travelling to Campbell's office without a story idea, he opened a volume of Gilbert and Sullivan at random to an illustration from *Iolanthe* showing the Fairy Queen at the feet of Private Willis of the Grenadier Guards; soldiers suggested empire, empire suggested Rome, and Rome, transposed, became a Galactic Empire whose fall he would narrate in Gibbon's manner. Four things passed from Gibbon into the Trilogy. The first is duration: Gibbon's decline extends across thirteen centuries, from the Antonines to the fall of Constantinople, and Asimov's Empire is likewise dying for generations before it notices. The second is structural causation. Gibbon's "General Observations" attribute the fall to the weight of the edifice itself, in which prosperity ripened the principle of decay until "the stupendous fabric yielded to the pressure of its own weight"; Seldon's prognosis is of the same order, a fall by gravity rather than by blow. The third is Gibbon's irony toward religion, examined in Chapter IV. The fourth is a repertory of episodes. Belisarius, recalled and disgraced by Justinian, becomes Bel Riose under Cleon II; Gibbon's strong generals destroyed by weak emperors, Stilicho, executed on Honorius's orders in 408, and Aetius, killed by Valentinian III in person in 454, after which a courtier told the emperor he had cut off his right hand with his left, supply the logic of "The General"; the sack of the eternal city becomes the Great Sack of

Trantor; and the shrunken imperial remnant appears in the senile Emperor Dagobert IX on Neotrantor, still asserting universal sovereignty over a handful of worlds.

The last name opens a homology that criticism has overlooked. Dagobert is the name of the last effective Merovingian king, after whom the Frankish monarchy dissolved into the *rois fainéants*, figurehead kings governed by the Mayors of the Palace. The Foundation's chief magistrates are Mayors. In the Frankish case, a palace mayoralty became a dynasty and then, with Charlemagne's coronation in 800, a renewed empire, whose palace school and scriptoria preserved a large share of the classical Latin literature that survives. The Foundation's arc, from the mayoralty of a scholarly settlement to the presumptive core of a Second Empire, follows the Carolingian rather than the Byzantine line. The Trilogy's Galaxy is thus late antiquity split along the axis between Constantinople and Aachen: the Empire borrows its pathos from the long-lived East, the Foundation its destiny from the reborn West.

3.2 Montesquieu's General Causes and the Dead Hand

Montesquieu's *Considérations sur les causes de la grandeur des Romains et de leur décadence* (1734) contains, in its eighteenth chapter, the sentence that "The General" dramatises. In my translation: it is not Fortune that rules the world; there are general causes, moral or physical, which act in every monarchy, raising it, sustaining it, or casting it down; all accidents are subject to these causes, and if the chance of a battle, that is, a particular cause, has ruined a state, there was a general cause that made it necessary for that state to perish by a single battle. Originally titled "Dead Hand" (1945), the story stages the last great general of the Empire, Bel Riose, loyal and brilliant, against a Foundation whose agents scramble to stop him and fail. Their failure does not matter. As the Siwennese patrician Ducem Barr explains, a strong general under a weak emperor would turn his armies on the throne, while a strong general under a strong emperor, Cleon II, will be recalled before he can win; either way, the Foundation survives. The particular causes, Lathan Devers's intrigues included, are epiphenomenal; the dead hand of Seldon is Montesquieu's general cause given a proper name. This is the high-water mark of determinism in the Trilogy, and *Foundation and Empire* places it immediately before the Mule. The volume is a philosophical diptych: Montesquieu's general cause, then Maxwell's singular point.

3.3 Ibn Khaldun: ʿAsabiyya and the Three Generations

Ibn Khaldun's *Muqaddimah* (1377) announced a new science, *ʿilm al-ʿumrān*, the science of human social organisation, whose laws would allow the historian to test reports against the nature of society, and it remains the most rigorous premodern anticipation of a predictive social science. Its central variable is *ʿasabiyya*, group solidarity, forged in the hardship of desert

life, expended in conquest, and dissolved by sedentary luxury, so that a dynasty has a natural lifespan of about three generations, while the prestige of a house lasts four: the builder, the one who knew the builder, the imitator, and the destroyer. The Foundation's ʿasabiyya is forged in Terminus's scarcity and the frontier vigour of the Traders; it sediments into the plutocracy of the merchant princes; and the hereditary mayoralty of the Indburs exhausts it in precisely three generations, from a capable and ruthless founder to a fastidious grandson obsessed with protocol, under whom the Foundation falls to the Mule almost without resistance. Yet Asimov inverts the Khaldunian cycle. In the *Muqaddimah*, the vigorous periphery conquers the decadent centre; in the Trilogy, the centre's knowledge colonises the periphery's vigour, and the Foundation prevails not through solidarity but through the strategic management of its absence. The Mule then supplies the missing term: his emotional "conversion" of enemies into devotees is ʿasabiyya manufactured *ex nihilo*, group feeling produced by technique rather than by hardship.

3.4 Vico: The Barbarism of Reflection

Vico's *Scienza Nuova* (1725; definitive edition 1744) founded its claim to a new science on the principle that the civil world, having been made by human beings, has principles discoverable within the modifications of the human mind itself (§331). Its course of nations passes through ages of gods, heroes, and men, and ends in the "barbarism of reflection," in which an over-refined reason dissolves the common sense that binds a people, preparing the *ricorso*. The Empire's intellectual senescence is depicted in exactly these terms. Lord Dorwin, the Imperial Chancellor who visits Terminus in "The Encyclopedists," pursues the question of humanity's planet of origin solely by collating the opinions of earlier authorities, and finds the suggestion that one might go and look faintly vulgar; his five days of diplomatic conversation, submitted to symbolic-logical analysis, are found to contain no commitments whatsoever. This is a language that has become pure form and a scholarship that has become pure commentary, the barbarism of reflection in its Galactic phase.

The Trilogy also enacts a more intricate Vichian movement. Asimov initially inverts the *verum-factum* principle: Vico held the human world more knowable than nature because human beings made it, whereas psychohistory renders the human world knowable by treating it as nature, as a gas. The Second Foundation then reinstates Vico against Asimov. Its mentalics is a science of the mind's modifications, an inner science, and its mastery of history derives from making history, so that *verum et factum convertuntur* in the strongest possible sense: the Second Foundation knows history because it produces it. The Vichian ages can be read in the Foundation's sequence as well: an age of gods in the oracular Vault and the Church of Science; an age of heroes in the Traders and merchant princes; an age of men in the legal-rational bur-

eaucracy of the Indburs; and a *ricorso* in the Mule, a living man-god whose will is felt directly as emotion.

3.5 Spengler: The World-City and the Fellah

Spengler's *Der Untergang des Abendlandes* (1918; 1922) treats each high culture as an organism with a life course of roughly a millennium, in which living *Kultur* hardens into *Zivilisation*: the world-city, the petrified intellect, the rule of money, a late "second religiousness," and finally Caesarism, which breaks the dominion of money and of democracy, its political instrument. Trantor, a planet entirely enclosed in metal whose forty billion inhabitants exist to administer the Galaxy and who depend on food imported from agricultural worlds, is the Spenglerian world-city raised to planetary scale. The Church of Science on the Periphery is a second religiousness; the Mule, who ends the commercial and plutocratic hegemony of the Foundation and its Independent Traders by sheer personal power, is a Spenglerian Caesar; and Seldon's thousand-year interregnum is the span of one Spenglerian culture.

The most suggestive Spenglerian figure, however, is the *fellah*: the people who remain after a civilisation has died, living on its ruins outside history. After the Great Sack, Trantor's survivors become farmers tilling soil among the wreckage of the imperial metropolis. The First Speaker of the Second Foundation in the Trilogy's final movement is Preem Palver, a stout Trantorian farmer-trader travelling with his wife. The governor of history hides among those who have left history. The Second Foundation's address is the post-historical place itself: history's centre has moved to the site where, by every visible sign, history has ended.

3.6 Toynbee: Challenge, Response, Chrysalis

Arnold Toynbee's *A Study of History* (twelve volumes, 1934–61, six of them in print by 1939, abridged by D. C. Somervell in 1946) supplies the Trilogy's most complete conceptual grammar, whether or not Asimov consulted it directly. Toynbee's "stimulus of hard countries" describes Terminus, a world almost without metals, and the Foundation's response, the miniaturisation of nuclear technology, is a literal instance of what Toynbee called *etherialisation*: the transfer of energy to a higher sphere through the progressive simplification of apparatus, of which his examples included the passage from wired to wireless telegraphy. Toynbee's creative minority, which leads by attraction and mimesis, degenerates into a dominant minority, which rules by force and routine; the Encyclopedists become the Indburs. His external proletariat, the barbarians beyond the limes, is the Periphery's Four Kingdoms. His universal state is the Galactic Empire, and his universal church, the institution born in disintegration that serves as the chrysalis of the next civilisation, as Christianity bridged the Hellenic and the Western worlds, is the Church of Science, with one decisive inversion: Toynbee's churches grow from

below among the disinherited, whereas Asimov's is fabricated from above by a creative minority for export. Toynbee's rhythm of *withdrawal-and-return*, the creative figure's retreat and transfigured reappearance, structures the Trilogy as a whole, from Seldon's withdrawal into the Vault to the Second Foundation's concealment and eventual disclosure.

Toynbee's "schism in the soul" identifies four responses to disintegration, and all four are present: archaism in the Empire's antiquarian nostalgia; futurism in the Actionist party's militarism and in the Mule's revolutionary conquest; detachment in the Second Foundation's contemplative withdrawal; and transfiguration in the promised Second Empire, a secular kingdom of God. His later distinction (1948) between Zealot and Herodian responses to an encroaching civilisation fits the Periphery precisely: Askone, which forbids the Foundation's nuclear devices on religious grounds, is Zealot; Anacreon and Korell, which absorb them, are Herodian.

3.7 Polybius's Cycle and Braudel's Three Times

Polybius's sixth book describes the *anacyclosis* of constitutions, from kingship through tyranny, aristocracy, oligarchy, and democracy to mob rule and back to monarchy. The Foundation traverses a compressed anacyclosis: a technocratic aristocracy of Encyclopedist trustees; Hardin's founding mayoralty, quasi-monarchical under popular forms; a sacerdotal hegemony; a merchant oligarchy; a hereditary and tyrannical mayoralty; the Mule's monarchy; and a chastened republic. The difference is that the Plan converts the cycle from fate into instrument: each turn is not suffered but scheduled.

Fernand Braudel's *La Méditerranée* (1949) distinguished three temporalities: the nearly immobile time of geography, the slow rhythms of economies and societies, and the history of events, which he likened to surface foam borne on the tides. Psychohistory is a science of the *longue durée*; the Seldon Crises are conjunctures; the episodes are *événements*. The Trilogy narrates events while asserting that only the *longue durée* matters, and then produces an event, the Mule, that rearranges the *longue durée* itself. Braudel's hierarchy of times is inverted in a single figure.

3.8 The Periphery Preserves: Yavneh, Vivarium, and the Tacitean Secret

The negotiation between Seldon and the Commissioner Linge Chen has a precise structural antecedent. According to the Babylonian Talmud (Gittin 56b), Yohanan ben Zakkai, smuggled in a coffin out of Jerusalem, which Roman legions under Titus would destroy together with its Temple in 70, asked the Roman commander for "Yavneh and its sages," founding the academy that transformed a Temple religion into a textual civilisation able to survive dispersion. The pattern is identical: on the eve of catastrophe, a scholar bargains with imperial power for a peripheral refuge of learning, and the refuge becomes the matrix of a civilisation. Cassiodorus, re-

tiring from the service of a collapsing Ostrogothic administration to found the monastery of Vivarium in Calabria in the mid-sixth century, and the monastic scriptoria of Ireland and Northumbria at the western edge of the Roman world, repeat the topos of a periphery that preserves the knowledge of the centre. Asimov, the son of Jewish immigrants from Petrovichi and a declared secular humanist, need not have had Yavneh in mind for the homology to illuminate his design. The Foundation is a people of the Book in exile, and its *Encyclopedia* is a Mishnah of the physical sciences.

Tacitus records that the civil wars of 69 divulged the secret of empire: that an emperor could be made elsewhere than at Rome. The Trilogy's political geography generalises this arcanum. Power is made on Terminus, on Kalgan, on the worlds of the Periphery, anywhere but Trantor. The final disclosure then turns Tacitus inside out. The true power of the Galaxy resides at Trantor after all, but only because the centre, emptied of empire, has become the perfect hiding place of government.

3.9 Leviathan without Covenant

Thomas Hobbes's *Leviathan* (1651), written in the aftermath of civil war, describes the condition in which there is no common power: a war of every man against every man, without industry, arts, or letters, in which life is "solitary, poor, nasty, brutish, and short." Seldon's thirty-thousand-year interregnum is the Hobbesian state of nature projected onto a galaxy, and the absence of letters is precisely what the *Encyclopedia* is founded to forestall. The engraved frontispiece of *Leviathan*, attributed to Abraham Bosse, shows the sovereign's body composed of hundreds of small human figures, the multitude rendered visible as a single artificial person, a "mortal god." Psychohistory's human conglomerate is the same aggregated body seen through statistics rather than through law.

The decisive difference lies in how the sovereign is generated. Hobbes's sovereign is authorised by covenant, each person agreeing with every other to transfer the right of governing, and it stands visibly before those who authorised it. Seldon's sovereign is generated by computation, authorised by no one, and invisible to those it governs. The Trilogy offers an exit from the state of nature without a social contract: Leviathan without covenant. Chapter XII returns to this absence.

CHAPTER

IV

The Instrumental Sacred

Religion, Commerce, and the Ladder of Hegemony

> *"The various modes of worship which prevailed in the Roman world were all considered by the people as equally true; by the philosopher as equally false; and by the magistrate as equally useful."*
>
> Gibbon, *Decline and Fall*, ch. 2

4.1 Hardin as Gibbon's Magistrate

In "The Mayors," Salvor Hardin governs the barbarian Four Kingdoms through a religion of his own manufacture. Periphery priests trained on Terminus in the cult of the Galactic Spirit maintain the nuclear installations of their kingdoms as sacraments they do not understand; the Foundation's technology becomes holy, and its interdict becomes a weapon. When the Regent Wienis of Anacreon attempts to turn a refurbished Imperial battle cruiser against Terminus, the faith of the crew decides the issue, and the regent's coup collapses. Hardin occupies exactly the position of Gibbon's magistrate: the religion is to him neither true nor false, only useful. His maxim, "Violence is the last refuge of the incompetent," has two faces. It states an ethic of non-violence, and it names a technique of domination that substitutes belief for force. The Church of Science is the first systematic fictional demonstration of what Arthur C. Clarke would codify in 1973 as his third law, that sufficiently advanced technology is indistinguishable from magic, with the difference that Asimov, three decades earlier, treated that indistinguishability as a political resource to be cultivated. "The Traders" already marks the next transition: Limmar Ponyets frees an imprisoned agent on Askone by building a transmuter that turns iron into gold and recording a councillor's sacrilegious use of it, so that the sacred becomes leverage in the hands of a merchant.

4.2 Machiavelli's Numa and the Return to Beginnings

Machiavelli's *Discourses on Livy* (I.11) present Numa Pompilius, who, finding a ferocious people, turned to religion to make them governable and feigned counsel from a nymph to lend his ordinances authority; the next chapter warns that the decline of a state's religion is the surest sign of its ruin. Hardin is Numa. More striking is the third book's first chapter, which holds that republics and sects endure only if periodically drawn back to their beginnings, *ridurre ai principi*, through dramatic acts that renew the founding terror and virtue. The scheduled apparitions of the Time Vault are an institutionalised *ridurre ai principi*: at intervals the Founder returns, in image, to reconvene the community around its origin. Their efficacy lies not in content but in convocation. When, at the crisis of the Mule, the image discourses calmly on a civil war that is not taking place, the ritual collapses into a revelation of its own mechanism, and the Foundation's morale collapses with it. Machiavelli's warning about the decline of religion is borne out in a displaced form: the Foundation's sacerdotal hegemony is already exhausted by Mallow's time, and the Foundation survives only because the Plan has substituted another bond.

4.3 Saint-Simon's Council of Newton and Comte's Spatialised Stages

In the *Lettres d'un habitant de Genève* (1803), Henri de Saint-Simon proposed a Council of Newton: twenty-one savants and artists, elected by public subscription, who would hold spiritual power, direct the progress of humanity, and preside over temples dedicated to Newton. The scientists would be the clergy of a new faith; his *Nouveau Christianisme* (1825) extended the design. Auguste Comte, his former secretary, formulated the law of three stages, theological, metaphysical, and positive, in the *Cours de philosophie positive* (1830–42), and in the *Système de politique positive* (1851–54) designed a Religion of Humanity in which a priesthood of savants holds spiritual power and an industrial patriciate temporal power. The Foundation's institutional settlement is exactly this one: a priesthood of science and a patriciate of commerce.

The Trilogy, however, performs an operation on Comte that has gone unremarked: it *spatialises* the law of three stages. In Comte the stages succeed one another in time; in the Foundation they coexist in space. Positive science reigns at the core among Terminus's engineers; theological belief at the periphery among the Anacreonian priests; and a metaphysical intermediary, the doctrine of the Galactic Spirit, abstracts the technical into the numinous. The law of three stages becomes a law of three zones, an esoteric and exoteric geography of knowledge that may be called concentric Comteanism. Because the zones are produced by policy rather than by development, the core acquires an interest in keeping the periphery theological. Progress becomes a differential to be administered.

4.4 Bacon's Merchants of Light

The Baconian utopia had already joined science, commerce, and secrecy. In the *New Atlantis* (1627), Salomon's House sends twelve Fellows abroad under the names of other nations to bring back books, abstracts, and patterns of experiments; they are called Merchants of Light. The Fellows also deliberate which of their inventions to publish, take an oath of secrecy concerning those they judge fit to conceal, and reveal some to the State and some not. The Foundation's Traders are Bacon's Merchants of Light reversed: they carry light outward rather than home, and they sell it. Salomon's House, deciding which knowledge the State shall possess, is the Second Foundation's first ancestor in the literature of utopia. What Bacon fused in one institution, Asimov separates into three, the Traders, the Church, and the Second Foundation, and distributes across centuries.

4.5 Ulfilas and the Conversion of the Barbarians

Gibbon's thirty-seventh chapter narrates the conversion of the barbarians, and especially the work of Ulfilas, bishop of the Goths, who devised a Gothic alphabet and translated the scriptures, bringing Arian Christianity to a people it simultaneously bound to Rome and divided from it. The Foundation's training of Periphery clergy on Terminus is the Gothic mission rationalised: the barbarian kingdoms receive from the centre a religion that incorporates them while keeping them subordinate, the priests' exclusion from the scientific core playing the role that the Arian–Catholic divide played for the Goths. The fate of Jord Parma, a Foundation missionary whom Hober Mallow surrenders to a Korellian mob, marks the end of the missionary age.

4.6 From Sacrament to Commodity: Mallow and the Gentle Commerce

In "The Merchant Princes," Hober Mallow discovers that the sacerdotal strategy has reached its limit. Korell's Commdor bans missionaries; the Imperial remnant at Siwenna still runs nuclear power but can no longer understand or maintain it; and when a fugitive missionary seeks refuge aboard Mallow's ship, Mallow surrenders him rather than furnish Korell with a pretext for war. Tried for this on Terminus, Mallow prevails, becomes mayor, and defeats Korell by declining battle: its elite and its households have come to depend on Foundation appliances, and the war collapses with the economy. Mallow's surrender of the missionary is the purest instance in the Trilogy of what Weber ("Politics as a Vocation," 1919) called the ethic of responsibility, set against the ethic of conviction: he accepts the death of an innocent to avert a war.

The philosophical genealogy of Mallow's victory is the eighteenth-century doctrine of *doux commerce*. Montesquieu's *De l'esprit des lois* (XX.1) holds, in my translation, that it is almost a

general rule that wherever mores are gentle there is commerce, and wherever there is commerce mores are gentle. Kant's *Perpetual Peace* (1795) observes that the spirit of commerce cannot coexist with war and sooner or later takes hold of every people. Albert Hirschman's *The Passions and the Interests* (1977) reconstructed this doctrine as the ideological prehistory of capitalism, in which interests were enlisted to tame the passions. Mallow is the doctrine in action and also its weaponisation: commerce pacifies by creating dependence. The Korellian elite's appetite for status goods is Veblen's conspicuous consumption (1899), and the domestic sphere, the Commdor's household, becomes the front line.

The deepest analogue is Marx. To explain the mysterious character of the commodity in the first volume of *Capital* (1867), Marx had to seek an analogy in "the mist-enveloped regions of the religious world." The Trilogy narrates as history what Marx used as metaphor. The same objects, the Foundation's nuclear devices, pass from altar to market: first sacred fetishes, then commodity fetishes. The fetish changes its liturgy. Mallow's discovery is that the commodity is the more efficient sacrament, binding by desire rather than by awe, its worshippers recruited as consumers rather than converts.

4.7 Weber: Charisma, Routine, Re-enchantment

Weber's typology of authority in *Wirtschaft und Gesellschaft* (1922) holds that charismatic authority is unstable and must be routinised into traditional or legal-rational forms. The Foundation exhibits the full sequence: Hardin's and Mallow's charisma is routinised into the hereditary mayoralty, a traditional authority, administered by the Indbur bureaucracy, a legal-rational one. Weber's disenchantment of the world describes the Foundation's core; its periphery is deliberately re-enchanted. This is re-enchantment for export. In "Science as a Vocation" (1919), Weber cited Tolstoy's question, what shall we do and how shall we live, as the question science cannot answer. Psychohistory claims to answer Tolstoy's question for the Galaxy while leaving it unanswered for every inhabitant, and this asymmetry produces the Trilogy's characteristic moral idiom: a cool, ironic, instrumental pragmatism in which ethical judgment is reduced to the recognition of what the Plan permits.

4.8 Elkins Revisited: Saint-Simon, Not Marx

Elkins argued that Asimov distorts Marxism into cyclical elitism. The diagnosis of elitism stands, but the genealogy requires correction. The Foundation's stadial ladder of science, religion, commerce, plutocracy, conquest, and covert rule is not Marx distorted but Saint-Simon restored. Its subject of history is not a class but an expertise; its politics are technocratic rather than proletarian. Engels credited Saint-Simon with the vision that the government of persons would give way to the administration of things (*Socialism: Utopian and Scientific*, 1880). The

Second Foundation realises a darker inversion of that formula: the administration of persons as things, molecules in a gas whose distribution is managed. The proximate ancestry of psychohistory is accordingly American as well as European. It runs from the Council of Newton through positivism to Thorstein Veblen's call for a "soviet of technicians" in *The Engineers and the Price System* (1921) and to the Technocracy movement that swept the United States in 1932–33, the milieu of Asimov's adolescence.

4.9 War by Other Means

The Foundation's victories without battle realise an ancient ideal. Sun Tzu's *Art of War*, in Lionel Giles's translation of 1910, holds that supreme excellence consists in breaking the enemy's resistance without fighting, and Hardin and Mallow win exactly so: the Anacreonian coup collapses without a battle, and Korell's war ends without one. Clausewitz's *Vom Kriege* (1832) made war the continuation of policy by other means. The Foundation inverts the formula, since its religion and its commerce are war continued by other means. Michel Foucault proposed this inversion explicitly in his lectures of 1975–76, "Society Must Be Defended," arguing that political power reinscribes the relations of force established in conflict within institutions, economies, and languages. Read through Foucault, Hardin's maxim on violence is not a renunciation of force but a relocation of it. The war continues in sacraments and markets, where it is less visible and therefore more durable.

CHAPTER

V

The Cunning of the Plan

Crisis, Necessity, and the Paradox of the Protagonist

> Freiheit ist die Einsicht in die Notwendigkeit. *Freedom is insight into necessity.*
>
> Engels, *Anti-Dühring*, 1878, after Hegel

5.1 The Anatomy of a Seldon Crisis

A Seldon Crisis is a conjuncture at which the convergence of structural forces reduces the Foundation's alternatives to a single viable course. Hardin formulates the principle in "The Mayors": when a crisis arrives, the correct action is evident because it is the only one remaining, and statecraft consists in waiting until the alternatives have closed. In the language of dynamical systems, the Plan defines a basin of attraction and the Crisis is a narrowing of that basin into a single channel, a funnel through which every surviving trajectory must pass. In the language of games, it is a forced move. In the language of poetics, it is *anagnorisis*: the climax of each episode is not a deed but a recognition, the moment at which the protagonist perceives the necessity that has been closing around him. Aristotle placed recognition at the summit of tragic plot; the Trilogy makes it the summit of statecraft. The Seldon Crisis is anagnorisis converted into a method of government.

The Plan's self-description is, moreover, a theodicy in the precise Leibnizian sense. Leibniz's *Théodicée* appeared in 1710, the same year as Arbuthnot's argument from the birth ratio, and argued that God chose the best among compossible worlds, the evils of the actual world being the minimum compatible with the greatest good. Seldon chooses, among computable histories, the one that reduces the interregnum from thirty thousand years to one thousand; the millennium of suffering is justified as the least evil compatible with the best attainable outcome. The Foundation's citizens, confident that Seldon will provide, are Panglossian, and the final page supplies a Voltairean irony few readers have noticed: the First Speaker, master of the Plan, turns out to be a Trantorian farmer, a man who in Voltaire's sense cultivates his garden among the ruins.

5.2 The Cunning of Reason and the Invisible Hand with Fingers

Hegel's lectures on the philosophy of history describe the cunning of reason (*List der Vernunft*): reason sets the passions to work on its behalf, so that world-historical individuals pursue particular aims and accomplish the universal without intending it. The Plan employs exactly this cunning, harnessing Hardin's political ambition, Mallow's acquisitiveness, the greed of barbarian kings, and the vanity of the Commdor. Its earlier ancestors are Mandeville's *Fable of the Bees* (1714), in which private vices yield public benefits, and Adam Smith's invisible hand (*Theory of Moral Sentiments*, 1759; *Wealth of Nations*, 1776), by which individuals promote ends that formed no part of their intention.

The Trilogy literalises the metaphor. What in Smith is emergent becomes in Asimov an organisation: the invisible hand acquires fingers. Adam Ferguson's *Essay on the History of Civil Society* (1767) gave the Scottish Enlightenment its formula for spontaneous order, establishments that are "the result of human action, but not the execution of any human design." The Trilogy reverses Ferguson exactly. What appears to be the result of human action but not of design turns out to be the execution of a hidden design. The great Enlightenment distinction between spontaneous order and deliberate construction, which Hayek would make the foundation of his social philosophy, collapses: the market and the Plan are revealed as one mechanism seen from inside and outside the cabinet.

5.3 Freedom as Insight into Necessity

The formula that Engels attributed to Hegel descends from Spinoza, for whom freedom consists in adequate understanding of the causes that determine us. Its narrative embodiment is Hardin's refusal of the Actionist party, whose young leader Sef Sermak demands armament and decisive action; Hardin's wisdom is the wisdom of waiting. Tolstoy's Kutuzov, who defeats Napoleon by declining to force events, is his literary prototype. Yet the Trilogy imposes on the Engelsian formula a decisive twist. For Engels, insight into necessity grants humanity collective mastery over nature and society. In the Trilogy, that insight is withheld from the many and reserved for the few, so that freedom becomes a privilege of epistemic position. Those who comprehend necessity, the Speakers, are free; those who embody it, the Foundationers, are merely fortunate. The formula of emancipation becomes the charter of a caste.

5.4 Virtù, Fortuna, and the Manufactured Occasion

The twenty-fifth chapter of *The Prince* likens fortune to a violent river and virtù to the dikes and embankments built in calm weather; the sixth credits Moses, Cyrus, Romulus, and Theseus with the virtue to seize an *occasione* that supplied the matter for their form. The Sel-

don Crisis is a manufactured occasion. The Plan arranges that the matter is ready when the man arrives, so that the Foundation's statesmen exercise a virtù of recognition rather than of creation. Psychohistory does not abolish fortune; it domesticates fortune statistically, converting chance into variance around a computed mean. The Mule is fortune's return as a flood the dikes were never designed to hold.

5.5 The Paradox of the Protagonist

The Trilogy's central formal contradiction is that a metaphysics of the mass is narrated through a gallery of clever individuals. Elkins observed that the stories celebrate individual cunning while the theory denies its efficacy. Georgi Plekhanov's *On the Question of the Individual's Role in History* (1898) offered the classical reconciliation: individuals act as the instruments through which social necessity operates, and they are substitutable, so that had a stray bullet killed Bonaparte, another general would have filled his place. The Trilogy formally assents to Plekhanov. Hardin and Mallow are, in principle, whoever happened to be there. But the narrative compels the reader to care about them as singular, and that compulsion is not incidental.

The novel is the genre of the irreplaceable individual. Lukács's *Theory of the Novel* (1916) defined it as the epic of a world abandoned by God, centred on a problematic individual in search of meaning; Ian Watt's *The Rise of the Novel* (1957) grounded its formal realism in the individualism of the modern subject. A novel about psychohistory is therefore a contradiction in genre: its form is an anti-psychohistorical technology, committed to the non-substitutability of persons that its content denies. The claim of this chapter is that the novel form is the Trilogy's internal Mule, the singular that the theory cannot assimilate, which the arrival of the Mule merely externalises. When the mutant appears, the content catches up with the form: the irreplaceable individual whom the narrative had always privileged against its own theory becomes that theory's falsifier.

The Trilogy's episodic construction confirms the point. Its narratives are separated by gaps of decades, and those ellipses are where the masses live: the statistical averaging that psychohistory performs on quadrillions of lives is performed by the text on the intervals between its stories. Psychohistory resides in the blank spaces between chapters; individuals reside in the chapters. The Trilogy is a novel whose theory is written in its margins and refuted on its pages.

5.6 The Alibi of Necessity

Isaiah Berlin's "Historical Inevitability," delivered in 1953 as the Auguste Comte Memorial Lecture and published in 1954, argued that doctrines of historical inevitability, providential or scientific, dissolve moral responsibility, turning judgments of praise and blame into category mistakes and supplying alibis to those who commit or tolerate wrongs in history's name. The

Trilogy's moral idiom demonstrates Berlin's charge. Mallow's surrender of the missionary, the Second Foundation's account of the fifty, and the Plan's justification of a millennium of suffering all invoke necessity: no other course was available. Considered morally, the Seldon Crisis is a device for manufacturing alibis, because a crisis is defined as the moment when no choice remains, and where no choice remains there appears to be no one to blame. The Trilogy itself knows better. A crisis that has been designed has a designer. The responsibility the crisis seems to dissolve is relocated to whoever arranged it, which is why the identity and location of the arranger must be hidden.

CHAPTER

VI

The Amplified Singular

The Mule as Mutation, Exception, and Ressentiment

> *"...twice two makes four is an excellent thing, but if we are to give everything its due, twice two makes five is sometimes a very charming thing too."*
>
> Dostoevsky, *Notes from Underground*, 1864, trans. Constance Garnett

6.1 The Event

In the fourth century of the Foundational Era, a conqueror known only as the Mule seizes Kalgan and begins absorbing worlds. The Foundation, now a hereditary plutocracy under Mayor Indbur III, anticipates a Seldon Crisis in the form of civil war with the Independent Traders and assembles in the Time Vault. The image of Seldon appears and discusses precisely that civil war, which is not occurring. For the first time the oracle is wrong. The Mule's power is the adjustment of emotion: he instils despair in enemies and devotion in converts such as Captain Han Pritcher. He is emaciated, beak-nosed, physically grotesque; he travels in disguise as Magnifico Giganticus, a clown who plays the Visi-Sonor, an instrument that produces emotion through sound and light; and he is sterile, whence his name. Bayta Darell, who has befriended the clown, understands on Trantor who he is at the moment when the psychologist Ebling Mis, driven to exhaustion by the Mule's covert stimulation, is about to name the location of the Second Foundation. She shoots Mis. The Mule had never touched her mind, because he prized the one uncoerced affection he had known.

6.2 The Quantum Gene and the Amplifier

Erwin Schrödinger's *What Is Life?* (1944), published the year before "The Mule," argued that the gene is an aperiodic crystal and that mutations are quantum jumps between stable molecular configurations. A mutation is therefore a quantum event preserved and amplified by an organism. Pascual Jordan had argued in the 1930s that living things possess control centres in

which quantum indeterminacies are amplified into macroscopic behaviour, so that organic action escapes classical determinism. The Mule is Jordan's amplifier at Galactic scale: a quantum event in a single genome, amplified by a nervous system able to amplify emotion in others, amplified by conquest into the destiny of a galaxy. Psychohistory is a classical statistical mechanics, and the Mule is the channel through which quantum indeterminacy enters it, the very channel through which twentieth-century physics breached Laplacean determinism. He is also a Maxwellian singular point (Chapter II): not a large cause but a small one situated at an instability, where an infinitesimal variation yields a finite divergence of history. The genre had supplied the figure of the mutant superman in Olaf Stapledon's *Odd John* (1935) and A. E. van Vogt's *Slan*, serialised in *Astounding* in 1940; Asimov supplied its epistemology, the mutant as the event outside the model.

6.3 The Exception and the Miracle

Carl Schmitt's *Political Theology* (1922) opens with the definition that the sovereign is he who decides on the exception, and argues that the exception in jurisprudence is analogous to the miracle in theology, while the deist God who does not intervene in the world corresponds to the liberal constitutional state that excludes the exception from its order. The correspondence with the Trilogy is exact. The Plan is the norm, a deistic order of general laws. Seldon is the deist creator who has withdrawn from his creation and appears in it only as a recording. The Mule is the miracle, an intervention from outside the order's laws. The First Foundation, a liberal-plutocratic order of norms, possesses no organ for deciding on the exception and collapses without a battle. The Second Foundation exists precisely as that organ. Giorgio Agamben's *State of Exception* (2003) argued that in modern governance the exception has become the paradigm of rule, and the Second Foundation, after the Mule, governs permanently in the exceptional mode. The Mule does not merely interrupt the Plan; he reveals what the Plan always required, which is a sovereign.

6.4 The Great Man Controversy Resolved

Thomas Carlyle's *On Heroes, Hero-Worship, and the Heroic in History* (1841) made history the biography of great men; Herbert Spencer's *The Study of Sociology* (1873) replied that the great man is the product of his society and must be explained by it; Plekhanov argued that great men are substitutable. The decisive intervention in this controversy for reading the Trilogy is William James's "Great Men and Their Environment" (1880), which answered Spencer by analogy with Darwin: great men are like spontaneous variations, whose causes lie in the physiological realm beneath social explanation, while the environment merely selects among them. The Mule is James's great man literalised as a biological mutation. He falsifies Plekhanov's substitu-

tion thesis, since no one else could have done what he did, and vindicates James: the decisive historical individual originates outside the social order and is selected, not produced, by it.

6.5 The Underground Man against Buckle

Notes from Underground (1864) is an explicit polemic against Buckle's statistical determinism and against the rationalist utopia of Chernyshevsky's *What Is to Be Done?* (1863), with its Crystal Palace. The Underground Man insists that human beings will act against their own advantage simply to prove they are not piano keys, will go mad deliberately if reason leaves them no other refuge, and will cherish twice two makes five. He is sick, spiteful, ugly, humiliated. The Mule is the Underground Man empowered: a deformed and mocked figure whose conquest of the Galaxy is the revenge of spite on the table of logarithms. The kinship is not merely temperamental. Dostoevsky aimed his polemic at the tradition of Quetelet and Buckle, the same tradition psychohistory perfects. Nietzsche's analysis of *ressentiment* in the *Genealogy of Morals* (1887, I.10) and Max Scheler's *Das Ressentiment im Aufbau der Moralen* (1912) supply the affective logic: the Mule's will to power is reactive, a creative revaluation born of injury.

6.6 The Master Who Cannot Be Loved

In the *Phenomenology of Spirit* (1807), the master's victory over the bondsman proves empty, because recognition from one who is not free is worthless; Kojève's Paris lectures (1933–39, published 1947) made this impasse the motor of history. The Mule is the Hegelian master raised to the absolute. Able to produce any emotion in anyone, loyalty, love, despair, he can receive none, since every affection he encounters may be his own product. Bayta's regard for Magnifico, formed toward a clown and unmanipulated, is the only recognition in the Galaxy he knows to be free, and he preserves it by refusing to enter her mind. That restraint, his single concession to the need for recognition, is precisely the opening through which she acts. The Mule is defeated neither by force nor by mentalics but by the structure of recognition itself: omnipotent affect cannot generate a free other. Kierkegaard's "Diary of a Seducer" (1843) had drawn the predicament a century earlier: Johannes engineers Cordelia's feelings so that she will seem to come to him freely, and the conquest, once achieved, is empty. This is the most profound philosophical moment in the Trilogy, and it belongs neither to the Plan nor to the Second Foundation but to Bayta, whose act unites pity and homicide in a single gesture. It is the only decisive act in the work performed outside both the Plan's computation and the Mule's control, and it makes her the Trilogy's ethical centre.

6.7 The Sterile Superman

Nietzsche's *Übermensch* is a bridge and a beginning. The Mule is an ending. His exception cannot propagate; his empire depends on his person; he is a Spenglerian Caesar who closes an epoch without founding one. His sterility is Asimov's biological guarantee that the exception will remain an exception: history heals at his death and the Plan can resume. Here lies the Trilogy's deepest conservatism. The singular is tolerable as a perturbation precisely because it cannot reproduce. The Second Foundation, by contrast, is a self-reproducing order, a lineage of mentalics transmitted by selection and training. Chapter VII will argue that the difference between the Mule and the Second Foundation is not one of power but of heredity: the Second Foundation is the Mule made fertile through institution.

6.8 The Clown, the Carnival, and the Aestheticisation of Politics

Mikhail Bakhtin's study of Rabelais (written 1940, published 1965) describes the carnival king, crowned and uncrowned in festive inversion, and carnival as a temporary suspension of hierarchy. Magnifico Giganticus is a carnival figure, and the Mule inverts carnival: the clown is crowned permanently, a carnival without Lent. Benjamin's essay on the work of art in the age of its technical reproducibility (1935–36) closes with the claim that fascism aestheticises politics. The Visi-Sonor is the aestheticisation of politics as an actual device: the Mule's music fills the Time Vault with despair, and the Foundation capitulates. Published in the year of Hitler's death, "The Mule" registers the charismatic mass-affect politics of the 1930s, the rally, the broadcast, the film, in the figure of an artist-conqueror who governs through mood. In Csicsery-Ronay's terms the Mule is the science-fictional grotesque, a confusion of the categories human and posthuman. In Iain M. Banks's later terminology (*Excession*, 1996), he is psychohistory's Outside Context Problem: an encounter for which the threatened order possesses no concepts at all.

CHAPTER

VII

The Dwarf in the Machine

The Second Foundation as Providential Machine

> *"Sir Isaac Newton and his followers have also a very odd opinion concerning the work of God. According to their doctrine, God Almighty wants to wind up his watch from time to time: otherwise it would cease to move."*
>
> Leibniz, first paper to Clarke, 1715, trans. Samuel Clarke, 1717

7.1 The Disclosure

Second Foundation (1953) unfolds in two movements. In "Search by the Mule," the conqueror, now ruling a Union of Worlds from Kalgan, hunts the Second Foundation with the converted Han Pritcher and the unconverted young Bail Channis, who proves to be a Second Foundation agent leading him toward a decoy. The First Speaker confronts the Mule and, at a moment when the conqueror's guard falters, reconfigures his mind so that he abandons the search and governs peacefully for the rest of his life. In "Search by the Foundation," scientists of the First Foundation led by Toran Darell, Bayta's son, detect an encephalographic signature of mental tampering, build a device that generates Mental Static, and, guided in part by the deductions of his fourteen-year-old daughter Arcadia, locate the Second Foundation at "Star's End," which they identify with Terminus itself. They capture some fifty of its members and believe it destroyed. A final chapter on Trantor discloses to a student that the fifty were volunteers sacrificed to produce exactly this belief, and that the Second Foundation has always been on Trantor, the solution turning on a proverbial echo of all roads leading to Rome. The Plan, the First Speaker explains, was never complete; it is a living structure maintained through the Prime Radiant, whose projected equations mark deviations for correction, and the Second Foundation envisions itself as the guiding stratum of the Second Empire to come.

7.2 Leibniz, Clarke, and the Reversal of Deism

In the correspondence of 1715–16, Leibniz mocked a Newtonian God obliged to mend his watch, a craftsman whose need to intervene proved his incompetence; a perfect God pre-establishes harmony and never needs to touch his creation. Samuel Clarke replied that God's continual governance is no defect but the mark of providence and dominion, and that a world running without him would be the world of atheism. Newton's *Opticks* had already envisaged a periodic divine reformation of the system as irregularities accumulate. The First Foundation's faith is Leibnizian: the Plan is a pre-established harmony that executes itself, and Seldon, having wound the clock, returns only to remind. The Second Foundation is Clarke's and Newton's God, correcting accumulated irregularities. Enlightenment theology travelled from Clarke's interventionist providence toward the non-interventionist order of the deists. The Trilogy travels the other way. It begins deist and ends theist, and in the vocabulary of secularisation it is a narrative of re-sacralisation: a clockwork galaxy discovers that it requires a watchmaker who never leaves the room.

7.3 The Providential Machine

Giorgio Agamben's *The Kingdom and the Glory* (2007) reconstructs the "providential machine" that Christian theology bequeathed to Western government. It has two poles: general providence, by which God governs through universal laws, and special providence, by which particular effects are executed through intermediate causes, above all through angels. The machine separates Kingdom from Government, reign from administration, in the formula that the king reigns but does not govern; it makes the angelic hierarchies the theological prototype of bureaucracy; and it requires Glory, the acclamation and liturgy that power needs even when it governs economically. The Trilogy maps onto this machine with remarkable precision. The Plan is general providence; the Speakers execute special providence; the First Foundation reigns visibly while the Second governs invisibly; and the apparitions in the Time Vault are Glory, the ceremonial acclamation of an absent sovereign whose prestige sustains the consent of the governed. The Greek *angelos* means messenger, one who speaks for another, and the Second Foundation's officers are called Speakers; its First Speaker is an archangel, and its Table of Speakers is Pseudo-Dionysius's celestial hierarchy reconstituted as an academic senate. The Second Foundation is an angelology.

7.4 Joachim of Fiore and the Order of Spiritual Men

Joachim of Fiore (c. 1135–1202) divided history into three *status*: the age of the Father, under law; the age of the Son, under grace; and an age of the Spirit, of contemplation and freedom, to

be inaugurated by a new order of spiritual men, *viri spirituales*, who would lead humanity into it. His followers dated its advent to 1260, and the Spiritual Franciscans understood themselves as that order. Löwith (*Meaning in History*, 1949) and Voegelin (*The New Science of Politics*, 1952) traced the triadic schemes of modern philosophies of history, Comte's three stages, Marx's three epochs, the Third Reich, to this Joachimite symbolism. The Trilogy is Joachimism equipped with a mathematical apparatus. Its triad runs from an age of law and necessity, the physical sciences of the First Foundation, through an age of mind, the mentalics of the Second, toward a Second Empire uniting them under the guidance of the spiritual order. The Second Foundation's mentalics are, word for word, spiritual men: men of mind. Its thousand-year interregnum is a millennium in the strict chiliastic sense of the twentieth chapter of Revelation.

7.5 Secularisation, Reoccupation, Gnosis

Three theories of modernity's relation to theology can be tested against the Trilogy. Löwith held that modern philosophies of history are secularised eschatology, the history of salvation without God. Hans Blumenberg (*Die Legitimität der Neuzeit*, 1966) countered that modernity did not secularise theological contents but "reoccupied" positions vacated by theology, answering questions, such as the meaning of history as a whole, that it had inherited without needing to pose. Voegelin diagnosed modern mass movements as Gnostic: they immanentise the Christian eschaton, and their elites claim a saving knowledge of history's structure. The Trilogy adjudicates the dispute by dividing it. The First Foundation confirms Blumenberg: psychohistory reoccupies the position of providence with statistical resources. The Second Foundation confirms Voegelin: it is a Gnostic elite in possession of a saving knowledge it does not share. What neither theory anticipates is the specific character of this gnosis. Because it is mathematical, it is communicable in principle; it is withheld not by its nature but by policy. The Second Foundation possesses a gnosis by decree, and Chapter VIII will show that the decree is not epistemically compelled.

7.6 Plato, the Grand Inquisitor, and the Hidden Righteous

Plato's *Republic* entrusts the city to guardians and philosopher-kings (473c–d) and prescribes a noble lie, the myth of the metals (414b–415d), to secure civic unity. The Second Foundation's founding fiction for the First, that the Second Foundation has been destroyed, is a noble lie in exactly this sense. Dostoevsky's Grand Inquisitor (*The Brothers Karamazov*, 1880) describes an elite that assumes the burden of knowledge and freedom so that the many may be happy, governing through miracle, mystery, and authority. The Trilogy supplies all three: miracle in Seldon's posthumous apparitions, mystery in the Second Foundation's unknown location, au-

thority in the Plan's prestige. The Inquisitor's self-image as a suffering elite matches the Speakers' self-image as servants who renounce the satisfactions of visible power.

The Jewish tradition of the *Lamed Vav*, the thirty-six hidden righteous in each generation by whose merit the world endures (Babylonian Talmud, Sanhedrin 97b; Sukkah 45b), offers a revealing contrast. The hidden righteous of later legend are unknown even to themselves, and they sustain the world without governing it. The Second Foundation is a Lamed Vav that governs: hidden righteousness converted into hidden rule.

7.7 The *Astounding* Unconscious: Wells, Hesse, Heinlein, Campbell

H. G. Wells imagined the Samurai of *A Modern Utopia* (1905), a voluntary ascetic order of rulers; *The Open Conspiracy* (1928), a conspiracy of the intelligent toward a world state, conducted openly; *The Shape of Things to Come* (1933), in which a technocratic elite rebuilds civilisation after collapse; and *World Brain* (1938), a permanent world encyclopaedia to coordinate human thought, the most direct ancestor of the *Encyclopedia Galactica*. The Second Foundation is Wells's Open Conspiracy closed. Hermann Hesse's *Das Glasperlenspiel* (1943), published the year after the first Foundation story, depicts Castalia, an intellectual province whose Glass Bead Game synthesises all knowledge in a quasi-musical mathematics. The Prime Radiant is a Glass Bead Game with executive power. But Castalia renounces power, and its Magister Ludi, Joseph Knecht, finally leaves the province because he grasps its historical fragility; the Second Foundation never leaves. Robert A. Heinlein's "Gulf," serialised in *Astounding* in November and December 1949 alongside the opening instalments of "—And Now You Don't," imagines a secret fraternity of cognitively superior "New Men" who covertly police the human future. In the same years Campbell's magazine was saturated with fantasies of mental supermen: van Vogt's *The World of Null-A* (1945), built on Korzybski's General Semantics; Hubbard's "Dianetics" (May 1950), which promised an exact science of mind; and the "psionics" Campbell championed through the following decade. The fantasy did not remain a fiction. Campbell soon broke with Hubbard, but in December 1953, the year the Trilogy's third volume appeared, Hubbard incorporated the Church of Scientology, an organisation built on his science of mind whose most advanced teachings are disclosed only to members who have completed its lower grades. Hardin manufactured a church out of physical science for barbarians; within four years of the first Dianetics article, Campbell's milieu had produced a church out of a purported science of mind, organised, like the Second Foundation, around graded esoteric knowledge. The Second Foundation is the *Astounding* unconscious: the collective fantasy of a mental science that would make its possessors the rightful governors of the species.

7.8 The Machine to Govern: Cybernetics and Requisite Variety

Norbert Wiener named his science after the Greek *kybernētēs*, steersman, and made feedback its principle of control (*Cybernetics*, 1948). Yet Wiener doubted that the social sciences could achieve such control, because their statistical runs are too short and their observers are coupled to what they observe. Reviewing *Cybernetics* in *Le Monde* in December 1948, the Dominican Dominique Dubarle imagined a *machine à gouverner*, a state apparatus run by a computing machine, and Wiener weighed the prospect warily in *The Human Use of Human Beings* (1950). The Second Foundation is the machine to govern built out of human beings: a feedback controller that measures deviations, displayed on the Prime Radiant, and applies corrections.

W. Ross Ashby, the builder of the homeostat (1948), later formulated the law of requisite variety (*An Introduction to Cybernetics*, 1956): only variety can absorb variety, so that a regulator must command at least as many states as the disturbances it is to counter. Ashby's law explains with formal precision why only a mentalic can defeat a mentalic. The Mule's disturbance has a dimension, the direct control of emotion, that no regulator built from physical science possesses; the Second Foundation alone has requisite variety. The law of requisite variety is the formal proof of the Trilogy's darkest implication.

7.9 The Distributed Mule

The technique by which the First Speaker defeats the Mule, entering a mind at a moment of weakness and reconfiguring its emotional dispositions, is the technique by which the Mule converted Pritcher. The two powers differ not in kind but in four accidents: visibility, since the Mule rules openly and the Second Foundation in secret; number, since he is one and they are many; legitimation, since his power serves ressentiment and theirs serves the Plan; and heredity, since he is sterile and they reproduce through selection and training. This is the Trilogy's suppressed equation. The Second Foundation is the Mule distributed, rendered invisible, legitimated by a theodicy, and made fertile through institution. Benjamin's emblem is thereby completed: the dwarf hidden in the automaton is also a hunchback, and the clown and the master of the game are one.

7.10 The Restoration of the End of History

Kojève's lectures read Hegel's end of history as the advent of a universal and homogeneous state, after which nothing fundamentally new can occur and a satisfied humanity ceases to act historically; Francis Fukuyama's "The End of History?" (1989) and *The End of History and the Last Man* (1992) revived the thesis for the order that followed the Cold War. The Galactic Empire at the Trilogy's opening is a universal and homogeneous state in Kojève's sense: one polity,

one species, no outside, a capital of administrators and a periphery of subjects. Its fall reopens history. The Plan's aim is not progress but the restoration of the end of history in the form of a Second Empire, and the Second Foundation is Kojève's Sage, the bearer of absolute knowledge who presides over a post-historical order. The Mule is history's return, and Bayta's act is its last free gesture before its administrators close it again. Read thus, the Trilogy is a restorationist epic whose happy ending is the end of the possibility of new beginnings.

CHAPTER

VIII

Arcana Probabilitatis

The Axiom of Ignorance and the Politics of Reflexive Prediction

> *"If men define situations as real, they are real in their consequences."*
>
> W. I. Thomas and Dorothy Swaine Thomas, *The Child in America*, 1928

8.1 The Axiom under Pressure

Psychohistory's second axiom requires that the predicted population remain ignorant of the prediction. *Second Foundation* dramatises the two inverse ways in which knowledge breaches the axiom. The first is complacency: the First Foundation's certainty that the Plan guarantees its triumph breeds a passivity that itself endangers the Plan. The second is revolt: the Darell circle's horror at being puppets breeds a resistance that endangers it equally. Both are effects of knowledge upon the known. The Second Foundation's closing stratagem is designed with exact economy against both. It restores the First Foundation's belief that the Plan holds while removing its knowledge that anyone holds the Plan: confidence without a governor, providence without a God. The purpose of the stratagem is to rebuild the conditions of the second axiom after they had been broken, and the fifty volunteers are the price of manufacturing a renewed ignorance.

8.2 Self-Fulfilling and Self-Defeating Prophecy

Robert K. Merton's "The Self-Fulfilling Prophecy" (*Antioch Review*, 1948), built on the Thomas theorem, described a false definition of a situation that evokes behaviour which makes the definition true, as when rumour of insolvency produces the run that ruins a sound bank, and recognised its inverse, the self-defeating prophecy. Popper called the influence of a prediction on the predicted event the Oedipus effect, after the oracle that brings about the parricide it foretells. Psychohistory's second axiom is the most radical conceivable response to reflexivity: it

eliminates reflexivity by eliminating knowledge. Another response was available, and it was published within four years of the Trilogy's completion.

8.3 The Fixed-Point Refutation

In 1954 Emile Grunberg and Franco Modigliani ("The Predictability of Social Events," *Journal of Political Economy*) and, independently, Herbert Simon ("Bandwagon and Underdog Effects and the Possibility of Election Predictions," *Public Opinion Quarterly*) proved, by way of Brouwer's fixed-point theorem, that if a population's reaction to a published forecast is a continuous function of that forecast over a bounded range, there exists at least one forecast that, once published, will be fulfilled despite, and indeed through, the population's reaction. A public, self-confirming prediction is possible. Applied to the Trilogy, the result is decisive. A psychohistory powerful enough to compute the Galaxy's reaction to any forecast could publish one of its fixed points and remain correct. The second axiom is not a mathematical necessity.

Why, then, secrecy? Because publication confines the attainable outcomes to the set of fixed points, whereas concealment leaves the controller free to reach any outcome its interventions can produce. Publication binds the predictor to futures that can be announced; secrecy enlarges the reachable set. The axiom of ignorance is the condition not of predictability but of *discretion*. This reframes the Second Foundation's concealment as a constitutional choice rather than an epistemic constraint: the choice of discretion over rule.

Finn Kydland and Edward Prescott's "Rules Rather than Discretion: The Inconsistency of Optimal Plans" (1977) showed that when agents anticipate policy, a policymaker who re-optimises at each moment produces worse outcomes than one credibly committed to a rule; the remedy, precommitment, is publicity's name in economics. The Seldon Plan oscillates between the two regimes. As a recorded, scheduled, publicly ritualised sequence of pronouncements, it is a rule. As the Second Foundation's covert adjustments, it is discretion. The Trilogy's arc moves from rule to discretion, and it hides the discretion in order to evade the time-inconsistency Kydland and Prescott would diagnose: a discretionary sovereign can be effective only if no one knows that it exists. Secrecy is the price of a discretionary sovereign's credibility.

8.4 Arcana Imperii

Tacitus reports that Tiberius sensed in a senatorial proposal an attempt on the *arcana imperii* (*Annals* 2.36), the secrets of rule. Early modern reason of state systematised the concept, from Giovanni Botero's *Della ragion di stato* (1589) to Arnold Clapmarius's *De arcanis rerumpublicarum* (1605), and Ernst Kantorowicz's "Mysteries of State" (1955) traced absolutist arcana to the transfer of ecclesiastical mysteries to the state. The axiom of ignorance is an *arcanum imperii* expressed in the idiom of methodology. Leo Strauss's *Persecution and the Art of Writing*

(1952) described philosophical esotericism, an exoteric teaching for the many and an esoteric one for the few. The Trilogy's political order is esoteric writing made into an institution. The First Foundation receives the exoteric teaching, a Plan, a destiny, a free civilisation; the Second Foundation keeps the esoteric one, which is that the civilisation is governed.

8.5 Performativity: An Engine, Not a Camera

Robert Lucas's critique (1976) showed that econometric models estimated under one policy regime fail under another, because agents' decision rules embed their expectations of policy. Goodhart's law (1975), generalised by Marilyn Strathern (1997), holds that a measure made into a target ceases to be a good measure. Donald MacKenzie's *An Engine, Not a Camera* (2006), building on Michel Callon, showed that financial models such as Black–Scholes–Merton reshaped markets to resemble the model. The Plan is an engine, not a camera: it does not describe the Galaxy's future, it produces it. J. L. Austin's *How to Do Things with Words* (1962) distinguished constative from performative utterances, and Seldon's pronouncements in the Vault are performatives disguised as constatives. The first apparition, which declares the *Encyclopedia* a pretext, dissolves the Encyclopedists' authority and opens the way to Hardin's government: it does what it says. The Trilogy's genuine science is not prediction but performativity, and the second axiom is the device that keeps the performance looking like a description.

8.6 Governmentality, Instrumentarian Power, Choice Architecture

Foucault's lectures *Security, Territory, Population* (1977–78) described apparatuses of security that act on the aleatory, on series of probable events, on populations known through statistics, regulating a milieu rather than commanding subjects; government becomes the conduct of conduct. Psychohistory is the apparatus of security scaled to a galaxy. Shoshana Zuboff's *The Age of Surveillance Capitalism* (2019) named "instrumentarian power," the capacity to know and shape behaviour at scale for others' ends, and located its foundation in a "division of learning," the question of who knows, who decides who knows, and who decides who decides. That division is the Trilogy's central axis, and the Second Foundation's power is a monopoly over it. Richard Thaler and Cass Sunstein's *Nudge* (2008) theorised choice architecture under the banner of libertarian paternalism. The Second Foundation's adjustments, minute displacements of emotional disposition that leave the subject's sense of choice intact, are choice architecture at the level of affect.

8.7 The Reader as Population

The Trilogy imposes the second axiom on its own readers. The location of the Second Foundation is withheld for two volumes; the reader is led through two false solutions, the decoy world of Tazenda and the apparent unmasking on Terminus, before the true one; and the pleasure of each Seldon Crisis depends on the reader's not having computed it. Narrative surprise is the aesthetic form of the second axiom. The epigraphs from the *Encyclopedia Galactica*, drawn, a prefatory note informs us, from an edition published in 1020 F.E., beyond the Plan's appointed term, place the narrating voice inside the fulfilled future: the reader reads from within the accomplished eschaton, as the reader of the Gospels reads after the Resurrection. Rereading converts the reader from population into Speaker, and herein lies the Trilogy's peculiar pedagogy and its peculiar seduction. To enjoy rereading the Trilogy is to enjoy the Speaker's pleasure, which is the pleasure of watching others not know. The work's enduring appeal to technocratic readers, examined in Chapter XI, is a function of this structure.

CHAPTER

IX

The Thermodynamics of Benevolence

Entropy, Information, and the Landauer Cost of History

> *"...a being whose faculties are so sharpened that he can follow every molecule in its course..."*
>
> Maxwell, *Theory of Heat*, 1871

9.1 Entropy as Master Metaphor

The Trilogy's governing image of decline is entropic. The Empire does not lose a war; it dissipates. Institutions lose structure, knowledge disperses, and technology degrades as the capacity to maintain it decays, so that the viceroyalty of Siwenna still runs nuclear plants that no one living can repair. Seldon's choice between thirty thousand years and one thousand is a choice between two rates of entropy production. The Empire is what Ilya Prigogine would call a dissipative structure, an order far from equilibrium sustained by continuous throughput, as Trantor is sustained by imported food, and its collapse is the failure of the flows that maintained it. Schrödinger had argued in 1944 that living things preserve their order by drawing negative entropy from their surroundings; the Foundation is precisely such an organism, growing as the Empire decays and feeding on its dissipation. The Plan cannot prevent the fall, which the second law forbids, but it can shape the dissipation and seed a new structure within it.

9.2 Maxwell's Demon and Its Exorcisms

Maxwell's thought experiment, sketched in a letter to Tait in 1867 and published in the *Theory of Heat* (1871), imagines a being who operates a trapdoor between two chambers of gas, admitting swift molecules in one direction and slow ones in the other, and so creates a temperature difference without expending work, in apparent violation of the second law. William Thomson

christened the being a demon in 1874. The century of exorcisms that followed is directly relevant to the Trilogy. Leo Szilard (1929) argued that the demon must measure, and that measurement carries an entropic cost. Léon Brillouin (1951; *Science and Information Theory*, 1956), building on Shannon (1948), identified information with negative entropy and argued that acquiring it costs at least as much entropy as it allows the demon to remove. Rolf Landauer (1961) showed that logically irreversible operations, above all the erasure of information, dissipate a minimum of $kT \ln 2$ of energy per bit. Charles Bennett (1982) completed the analysis: measurement can in principle be performed reversibly, and the irreducible cost lies in erasing the demon's memory so that it can begin the next cycle. The demon is reconciled with the second law not by the cost of perceiving but by the cost of forgetting.

9.3 The Second Foundation as Demon

The Second Foundation performs the demon's three operations. It *measures*, through mentalic perception of emotional states and the Prime Radiant's tracking of deviations. It *sorts*, through adjustments that allow some historical trajectories to pass and close the door on others. And it *resets its memory*, erasing the traces of its own operations so that the First Foundation remains statistically independent of it, which is to say, random with respect to its governor. On the Landauer–Bennett analysis, the first two operations can in principle approach zero cost. The third cannot. The Second Foundation's irreducible costs fall exactly where physics predicts they must: in erasure.

9.4 The Landauer Cost of History

I therefore propose a historical analogue of Landauer's principle. Every agency that reduces the entropy of a historical system by acting on information it conceals must pay an irreducible cost in erasure, and in human systems that cost takes the form of erased persons, rewritten minds, and falsified collective memory. The cost can be displaced, onto volunteers, onto the deceived, onto the future, but it cannot be abolished. The Trilogy's ledger can be reconstructed with some precision, and it must be kept in the active voice, because concealing its agents is how the Trilogy keeps these costs off the books. Bayta Darell kills Ebling Mis to erase the Second Foundation's location from a mind about to pronounce it. The First Speaker erases the Mule's will and installs in its place the disposition of a placid ruler. The Second Foundation's leadership gives up fifty volunteers from its own ranks to the First Foundation, whose agents seize them with a device that generates Mental Static, so that the Second Foundation may be erased from the First's knowledge. The Second Foundation overwrites the First Foundation's collective memory with the conviction of a victory that did not occur. The First Speaker, Preem Palver, compromises Arcadia Darell's autonomy, since the journey, the rescue, and the conclusions

she carries home prove to belong to his choreography. Finally, Asimov himself erases the reader's first innocence with a revelation that converts every earlier page into the record of a managed process.

The Trilogy never tallies this ledger. Its costs are absorbed in ellipses and closing revelations; the Second Foundation's last scene is a seminar, not a funeral. The ledger yields a general law that may be called the *Landauer limit of benevolence*: the more completely a hidden benevolence governs, the more it must erase, so that total concealed benevolent government converges on the total erasure of the governed's knowledge of their own history.

9.5 Asimov's Three Verdicts

Asimov's own fiction of the following six years delivers three verdicts on the Seldonian Demon. "The Evitable Conflict," published in *Astounding* in June 1950, five months after the last Foundation serial, imagines the Machines that run the world economy introducing small apparent errors that prove deliberate: members of an anti-Machine movement are quietly displaced into positions where they can do no harm, under a First Law generalised from individual human beings to humanity as a whole, and Susan Calvin concludes that humanity has lost its voice in its own future and is better for it. This is *affirmation*. *The End of Eternity* (1955) imagines an organisation outside time, Eternity, that performs Reality Changes of minimal magnitude to reduce human suffering; in eliminating risk it has eliminated space travel, so that humanity never reaches the stars and dwindles alone on Earth, until an emissary from the far future persuades the Technician Andrew Harlan to destroy Eternity and free humanity for the risks that will produce a Galactic Empire. This is *condemnation*, and its implication is remarkable: within Asimov's corpus, the Empire whose fall the Trilogy narrates is itself the fruit of the abolition of a Second Foundation. "The Last Question" (1956), which Asimov named his favourite among his stories, follows successive generations of computers asked whether entropy can be reversed, each replying that data are insufficient, until, after the heat death, the final intelligence, having absorbed every mind, finds the answer and speaks the opening command of Genesis. This is *apotheosis*: the negentropic fantasy terminates in God. Affirmation in 1950, condemnation in 1955, apotheosis in 1956: a single author delivers the full range of judgments on the benevolent hidden steerer.

9.6 Omelas and Arkanar

Ursula K. Le Guin's "The Ones Who Walk Away from Omelas" (1973) describes a city whose happiness depends on the misery of a single child confined in a cellar. The citizens know; most accept; some walk away. Omelas is the Landauer cost made public. In the Trilogy the cost is concealed; in Omelas it is a civic initiation, and Le Guin's provocation is that knowledge of the

cost is the condition of moral seriousness, since only those who know can walk away. The Second Foundation denies the governed even this option.

Arkady and Boris Strugatsky's *Hard to Be a God* (1964) sends Anton, under the identity of Don Rumata, from a communist Earth to observe a feudal kingdom, Arkanar, under a prohibition on intervention grounded in "basis theory," an exact science of historical stages descended from historical materialism. He watches the rise of a fascistic Holy Order and, after Kira, the woman he loves, is killed when the Order's men storm his house, breaks the prohibition in a massacre. The Strugatskys' Institute of Experimental History is a Soviet Second Foundation whose science forbids it to act, and whose agent disintegrates under the prohibition. Where the Second Foundation pays the Landauer cost in other people's lives, Rumata pays it in his own.

9.7 Active Forgetting and the Nation

Nietzsche's "On the Uses and Disadvantages of History for Life" (1874) argued that a living culture requires a *plastic power*, the capacity to heal wounds and to forget, and that an excess of historical knowledge paralyses. Ernest Renan's "Qu'est-ce qu'une nation?" (1882) held, in my translation, that forgetting, and even historical error, is an essential factor in the creation of a nation, which is why the progress of historical studies is often a danger to nationality. The First Foundation becomes the nucleus of a future empire by forgetting, and by believing an error about, its own governance. The Second Foundation administers this forgetting: it is a ministry of Nietzschean plasticity imposed from outside, a practice of *mnemopolitics*, the government of a civilisation's memory as the precondition of its health. The final irony is architectural. A civilisation founded as an *Encyclopedia*, a project of total memory, survives by managed amnesia.

9.8 The Forgotten Origin

The Galaxy of the Trilogy has forgotten where humanity began. Lord Dorwin's "Origin Question" is debated among Imperial scholars as a contest of authorities, and no one any longer knows which world first bore human beings. Plato's *Timaeus* (22b–23b) gives this amnesia its classical form: an Egyptian priest tells Solon that the Greeks are always children, because the periodic catastrophes that destroy their cities destroy their records, so that each generation begins again without memory of what came before. The Empire's forgetting of its origin is the macroscopic counterpart of the Foundation's managed amnesia, and the late sequels, in which references to Earth turn out to have been removed from the Galaxy's libraries, convert the Platonic catastrophe into an act. Mnemopolitics in the Trilogy's universe operates at every scale, from a civilisation's belief about its governors to a species' knowledge of its birthplace.

9.9 The Arithmetic of Persons

Asimov composed the Trilogy between 1942 and 1950, the years in which Nazi Germany murdered some six million European Jews and in which that genocide began to be documented. The Trilogy contains no genocide and no allegory of one; its catastrophe is decline, not extermination, and its governors aim to minimise suffering. Yet its characteristic gaze, which governs persons as molecules and history as a quantity, shares a structure that historians of the genocide identified among the conditions of its execution. Raul Hilberg's *The Destruction of the European Jews* (1961) reconstructed the genocide as an administrative process carried out through the ordinary machinery of a modern state, its ministries, railways, and accounts. Hannah Arendt's *Eichmann in Jerusalem* (1963) described Adolf Eichmann, the SS officer who organised the deportations to the killing centres, as a functionary whose crimes required not demonic hatred but thoughtlessness. Zygmunt Bauman's *Modernity and the Holocaust* (1989) argued that bureaucratic organisation produces moral indifference through distance and the division of labour, substituting technical for moral responsibility.

The point is not that the Second Foundation resembles the perpetrators; in its aims it is their opposite. The point is that the administration of persons as things (Chapter IV) is ethically indifferent in itself, and becomes benign only through the benevolence of the hand that applies it. Psychohistory contains no internal limit on that hand, no point at which the arithmetic must stop because a person stands in its way; the Trilogy's only safeguard is the stipulated goodness of the Speakers. A science that knows only *how many* cannot, from its own resources, refuse what is done to a *who*. Stanisław Lem's fictions on statistics and genocide, discussed in Chapter X, press the point from inside the genre, and the conclusion answers it with the principle of forgiveness, which is addressed only to persons.

CHAPTER

X

Comparative Galaxies

A Taxonomy of Responses to the Seldonian Demon

The speculative and philosophical fiction surrounding the Trilogy can be ordered by two variables: whether the governing order is legible, at least to an elite, or opaque to all; and whether hidden steering is endorsed or refused. Four modes result, and the Trilogy is the only major work to pass through all of them.

Mode	Legibility of the governing order	Stance toward hidden steering	Exemplary works	Figure of the steerer
Providential	Legible to an elite	Endorsed, with unease	Bacon, *New Atlantis*; Wells; Hesse, *Das Glasperlenspiel*; Heinlein, "Gulf"; Banks, the Culture novels	The guardian order
Apophatic	Opaque to all	Undecidable	Borges, "The Lottery in Babylon," "Tlön, Uqbar, Orbis Tertius"; Lem, "The New Cosmogony," *Solaris*; Calvino, *Invisible Cities*	The unknowable company
Insurgent	Made opaque by design	Refused	Dostoevsky; Zamyatin, *We*; Herbert, the *Dune* sequence; Le Guin; Dick	The unpredictable person or people
Tragic-progressor	Legible but unbearable	Endured as a burden	The Strugatskys; Miller, *A Canticle for Leibowitz*; Stapledon; Liu Cixin	The broken guardian

10.1 The Providential Mode

Bacon's Salomon's House, Wells's Samurai, Hesse's Castalia, and Heinlein's New Men (Chapters IV and VII) imagine guardian orders whose knowledge entitles them to steer, and each registers a characteristic unease: Bacon's secrecy oath, Wells's insistence that his conspiracy be open, Knecht's departure from Castalia. Iain M. Banks's Culture novels are the mode's most self-conscious modern form. The Culture is governed by Minds, vast artificial intelligences, and its covert wing, Special Circumstances, intervenes in less advanced civilisations. In *The Player of Games* (1988) the Empire of Azad organises its entire hierarchy around a game so

complex that it encodes the empire's ideology, and the Culture sends a champion player, Jernau Morat Gurgeh, whose victory, engineered by Minds that manipulate him as well, destabilises the empire. Banks revises the Trilogy in two respects. The Culture does not conceal its existence, only its interventions; and its manipulations extend to its own citizens, Gurgeh having been blackmailed into service. The Culture knows its Landauer cost, worries about it, and pays it anyway. Banks described his project as reclaiming space opera from its imperial and right-wing defaults, and the Culture is best read as a Second Foundation that has acquired a conscience without surrendering its methods.

10.2 The Apophatic Mode

Borges's "The Lottery in Babylon" (1941) narrates a lottery that expands until it governs every aspect of life, administered by a Company whose operations become silent and secret, until the Babylonians cannot decide whether the Company is eternal, governs all things, or has never existed, and the question ceases to matter. This is the end state of the Second Foundation: an institution so successful in concealing its operations that its existence becomes undecidable to those it governs. "Tlön, Uqbar, Orbis Tertius" (1940) describes a secret society of savants who, across generations, compose an encyclopaedia of an imaginary world that then begins to replace reality. Borges's encyclopaedia inverts Seldon's. Seldon's *Encyclopedia* was a decoy that became a monument; Tlön's was a fiction that became the world. Both join an encyclopaedia to a secret society, and both make the encyclopaedia an instrument of world-transformation. "The Garden of Forking Paths" (1941) imagines a labyrinth of time in which every possibility branches; the Seldon Crisis is its anti-garden, the point at which forking paths are made to converge.

Stanisław Lem's "The New Cosmogony," the fictional Nobel address that closes *A Perfect Vacuum* (1971), argues that the laws of physics are the stabilised outcome of a game played by ancient civilisations whose silence explains why the cosmos seems empty. Here the Second Foundation is scaled to cosmology: natural law is administration, and the governors' silence is part of their method. *Solaris* (1961) explores the opposite limit, an Other that no science can read, and *Golem XIV* (1981), a superintelligence lecturing humanity on the limits of its evolution, anticipates the substitution the Asimov apocrypha would make, an artificial mind in the governor's chair. Italo Calvino's *Invisible Cities* (1972) gives the apophatic mode its most melancholy image: Kublai Khan, sensing that his empire is an endless and formless ruin, searches Marco Polo's reports for a pattern that would make decay legible as design.

Two later fictions by Lem turn the apophatic form on the arithmetic of persons. "One Human Minute," the title piece of a 1986 English collection, reviews an invented statistical compendium that tabulates everything humanity does in a single minute, its births and deaths, its labour and its violence; on the reading offered here it mirrors the psychohistorical gaze, in

which the totalising view that makes the species legible renders its suffering weightless. *Provocation* (*Prowokacja*, 1984; in English as "The World as Holocaust") reviews an invented scholarly history of the Nazi genocide, bringing the fictitious review to bear on the century's central crime. Chapter IX has taken up the question both pieces put to the Trilogy.

10.3 The Insurgent Mode

Yevgeny Zamyatin's *We* (written 1920–21) depicts the One State's mathematically infallible happiness, regulated by a Table of Hours under an all-seeing Benefactor. Classical dystopia is total, *visible* planning. The Trilogy's innovation is invisible planning, and it transforms dystopia into a form that I call *cryptotopia*: a society whose utopian management is invisible to its members and is therefore experienced as freedom. The insurgent mode is the literature that refuses cryptotopia.

Aldous Huxley's *Brave New World* (1932) supplies the nearest dystopian likeness of the Second Foundation in its World Controller, Mustapha Mond, who keeps the suppressed science, scripture, and literature of the past and governs a population conditioned to contentment and tranquillised by soma: knowledge withheld and affect engineered, the two instruments of the Speakers. George Orwell's *Nineteen Eighty-Four*, published in June 1949, five months before the cover date of the first instalment of "—And Now You Don't," imagines the opposite technique for a kindred end: a Party that governs the future by rewriting the past in its records and by visible terror. Orwell's Inner Party and Asimov's Speakers both practise mnemopolitics, the one through a Ministry of Truth and a Thought Police, the other through an adjustment no one feels. Cryptotopia is the dystopia that has learned it no longer needs terror.

Frank Herbert's *Dune* (1965) and its sequels are the most sustained literary answer to the Trilogy. Paul Atreides's prescience is psychohistory internalised in a single mind, and it traps rather than liberates him: he foresees the jihad fought in his name and cannot prevent it. The Bene Gesserit breeding programme, pursued for millennia toward the Kwisatz Haderach, is a Second Foundation plan whose product arrives a generation early and escapes its makers, an engineered Mule. Herbert said he intended the series as a warning against charismatic leaders, and in *God Emperor of Dune* (1981) he turned that warning into an anti-Plan. Leto II imposes a tyranny of three and a half millennia, the Golden Path, designed to make humanity crave escape and to make it unpredictable; he breeds a line invisible to prescience, and after his death the Scattering disperses humanity beyond any single foresight. Seldon plans to restore predictability; Leto plans to abolish it for ever. Herbert's verdict is that the only plan worth executing is a plan to end planners.

Ursula K. Le Guin offers the mode's political alternative. *The Dispossessed* (1974) joins an anarchist critique of central planning to a theory of time, Shevek's unification of sequency and

simultaneity, that refuses both linear progress and pure cycle. Her Hainish Ekumen (*The Left Hand of Darkness*, 1969) is a galactic polity without empire, whose first envoy arrives alone and unarmed so that the encounter cannot be coercive: a non-sovereign galactic order, the antithesis of both the Galactic Empire and the Second Foundation. Philip K. Dick contributes two precise inversions. In "The Minority Report" (1956), a man's knowledge of his own predicted crime alters the prediction, the second axiom collapsing at the scale of a single life. In *Solar Lottery* (1955), a society organised by game theory selects its supreme ruler by randomised lottery and publicly appoints assassins to hunt him, so that randomisation becomes the defence against a predictable sovereign.

Two further works sharpen the insurgent case. Herbert's universe shares the Trilogy's closure condition in another key: after the Butlerian Jihad has outlawed thinking machines, computation and prediction are performed only by human minds, the Mentats and the prescient, so that, as in Asimov's robot-free Galaxy, the predictor must be human and its politics is the politics of a human caste. And in Dick's *The Man in the High Castle* (1962), set in a world where the Axis won the Second World War, the novelist Hawthorne Abendsen composes, with the help of the *I Ching*, a counter-history in which the Allies won. Prediction here produces not control but contingency: the oracle's work is to show that the actual is only one of the possible.

10.4 The Tragic-Progressor Mode

The Strugatskys' Noon cycle extends the tragedy of *Hard to Be a God*. In *Beetle in the Anthill* (1979), the security chief Rudolf Sikorski kills Lev Abalkin, possibly a sleeper agent of the vanished Wanderers, out of fear of an incalculable future, and the novel leaves the justice of the act unresolved. In *The Waves Extinguish the Wind* (1985–86), the Ludens, a new branch of humanity endowed with mental powers, emerge within humanity and depart from it, indifferent to its affairs: the Second Foundation's evolutionary destiny, a mentalic species that declines to govern. Walter M. Miller Jr.'s *A Canticle for Leibowitz* (1959) has monks preserve the Memorabilia of a civilisation destroyed by nuclear war, watch it rebuild and destroy itself again, and finally carry the faith to the stars. It inverts the Church of Science: in Asimov, science uses religion; in Miller, religion shelters science and cannot prevent its abuse. Both answer Gibbon's judgment that he had described the triumph of barbarism and religion. Olaf Stapledon's *Last and First Men* (1930) traces eighteen human species across two billion years and ends with the Last Men, facing extinction, scattering the seeds of life on a plan they will never see fulfilled; *Star Maker* (1937) shows a creator who contemplates his creatures aesthetically, without providence. Stapledon is the Trilogy's cosmic counterweight: history has no Plan, only a contemplation.

Liu Cixin's *The Dark Forest* (2008) completes the mode. Its cosmic sociology is deduced, like psychohistory, from two axioms, that survival is the primary need of a civilisation and that civilisations grow while the matter of the universe remains constant, supplemented by the concepts of chains of suspicion and technological explosion. Its theorem, however, is not a millennium of guided peace but universal predation. Its Wallfacer Project entrusts humanity's defence to individuals who may disclose their plans to no one, because the enemy can observe everything except thought, and so turns the second axiom against the planners' own side. Luo Ji's eventual deterrence makes him a solitary stochastic sovereign, holding his civilisation's survival on a threatened broadcast.

10.5 The Trilogy among Its Kin

The Trilogy is the only major work that occupies all four modes in sequence: providential in the self-executing Plan, insurgent in the Mule, tragic in Bayta's act, apophatic in a denouement that renders the Second Foundation undecidable to the civilisation it governs. Its taxonomic instability is its achievement. It contains its own critiques because it follows its premise far enough to meet them.

10.6 The Adaptation as Symptom

The television adaptation that began streaming in 2021 makes two revisions that function as diagnoses. Its Genetic Dynasty, an Empire ruled by successive clones of a single emperor, literalises Spenglerian petrification: history as a loop, sovereignty as a copy. And it reimagines Seldon as a persisting consciousness housed in the Prime Radiant and the Vault. The twenty-first century can no longer imagine the Seldonian Demon as a dead man's recording or a guild of psychologists; it imagines it as an artificial intelligence. The adaptation thereby completes the substitution the apocrypha began, robots behind the Plan, and leads directly to the present.

CHAPTER

XI

Seldonism

The Afterlife of Psychohistory and the Stochastic Sovereigns of the Present

> *"All actions relating to the right of other men are unjust if their maxim is not consistent with publicity."*
>
> Kant, *Perpetual Peace*, 1795, Appendix II, trans. M. Campbell Smith

11.1 From Fiction to Charter Myth

By *Seldonism* I designate the disposition in which the Trilogy functions not as a fiction to be interpreted but as a charter to be implemented. Its elements can be specified. Civilisational time is reckoned in millennia; collapse is treated as a probability whose duration can be shortened rather than as an event to be prevented; the preservation of knowledge is understood as insurance against a dark age; a small elite of quantitative foresight is regarded as the natural trustee of the species' future; and the ignorance of the governed is accepted as tolerable or even necessary. Seldonism is rarely professed as a doctrine. It is enacted as a sensibility, and its documentary traces are abundant.

Paul Krugman has said repeatedly that the figure of Hari Seldon drew him toward economics, a debt he acknowledged at length in 2012. Elon Musk has described the series as formative, and the Tesla Roadster launched into solar orbit on the first flight of the Falcon Heavy in February 2018 carried a quartz-glass disc, produced by the Arch Mission Foundation, on which the Trilogy was inscribed. The symbolism is exact: a private launch vehicle carried the Encyclopedists' story beyond the Earth as the token of a multiplanetary insurance policy against civilisational collapse. Peter Turchin, the founder of cliodynamics, published in *Nature* in 2010 a brief forecast that structural-demographic pressures would bring the United States a peak of political instability around 2020, and his *End Times* (2023) presents the turbulence of the intervening years as confirmation. Whether the forecast was confirmed remains contested. What matters here is its form: a published prediction entering the population it predicts, the precise situation the second axiom forbade. A forecast of instability that circulates among elites can

feed the anxieties it describes or provoke the reforms that falsify it, so that publicity contaminates verification in either direction. Cliodynamics lives in the regime the Trilogy declared impossible, and the Trilogy's prohibition has become an open empirical question.

11.2 The Archive Impulse

The first, ostensibly decoy function of Seldon's project, the *Encyclopedia*, has been literalised again and again. The Long Now Foundation, established in 1996, is building a clock designed to run for ten thousand years and has assembled a "Manual for Civilization," a library of books chosen to help rebuild a society after collapse. The Svalbard Global Seed Vault, opened in 2008, stores duplicate seed collections against agricultural catastrophe; the nearby Arctic World Archive (2017) received in 2020 GitHub's Arctic Code Vault, a snapshot of public open-source software sealed in a decommissioned mine; and an Arch Mission library travelled to the Moon aboard the Israeli lander Beresheet, which crashed there in April 2019. These are Terminuses: peripheral repositories founded against a dark age.

The Trilogy's own lesson, however, is that the *Encyclopedia* was the pretext. Archives preserve information but not the capacity to use it, as the viceroyalty of Siwenna, with its reactors that no one can repair, illustrates. What shortens a dark age in the Trilogy is not storage but institutions: living practices of knowledge, trade, and interpretation transmitted between persons. The archive impulse mistakes the decoy for the Plan. Its merit is that it is public; its limit is that a library without a community of readers is a tomb.

11.3 Longtermism as Secular Seldonism

The philosophical movement known as longtermism, articulated in Nick Bostrom's "Existential Risk Prevention as Global Priority" (2013), Toby Ord's *The Precipice* (2020), and William MacAskill's *What We Owe the Future* (2022), holds that because the potential future population is vast, improving humanity's long-term trajectory carries overwhelming expected value, so that reducing existential risk and preserving civilisation's capacity to recover from catastrophe become priorities. The structural homology with the Plan is close. The unit of moral concern is an aggregate of trillions extended in time, the human conglomerate projected forward; the instrument is a marginal shift in probabilities; the agents are a small community of forecasters, researchers, and funders; and the legitimacy is a computed minimisation of aggregate loss.

The Trilogy's two critical instruments apply. The Landauer question asks who pays the cost of steering, and expected-value reasoning over astronomically many potential persons can, if applied without constraint, license the displacement of costs onto the present, as the Plan licenses the sacrifice of the fifty. The publicity question asks whether steering that shapes the common future without a democratic mandate can be legitimate. Fairness requires recording

the movement's own answers. Its principal theorists insist on moral uncertainty, on pluralism, and on the priority of ordinary moral constraints over speculative calculation. The collapse of the FTX exchange in November 2022 brought the question into focus. Its founder, Sam Bankman-Fried, had been among the movement's most prominent funders, notably through the FTX Future Fund; in November 2023 a federal jury in New York convicted him on seven counts of fraud and conspiracy arising from the misappropriation of billions of dollars in customer deposits, and in March 2024 he was sentenced to twenty-five years in prison. William MacAskill and other leading figures publicly repudiated reasoning that treats expected future value as a licence for present wrongdoing. How far Bankman-Fried's conduct expressed longtermist reasoning, rather than exploiting its prestige, remains disputed; whatever its motives, the episode made the Landauer question concrete, since the costs of a venture justified by the long-term future fell on present persons who had not agreed to bear them. Defenders note, too, that reducing catastrophic risk is a goal shared far beyond the movement, and that much of its work is published and openly contested. The comparison with the Second Foundation is therefore a diagnostic rather than an accusation. Longtermism inherits the Plan's structure; whether it also inherits the Plan's secrecy is the question on which its legitimacy turns, and the openness of its published debates is its best defence against that inheritance.

11.4 Predictive Governance

The institutional apparatus of stochastic sovereignty has been assembled piecemeal. The British government's Behavioural Insights Team, founded in 2010 and widely called the "nudge unit," installed choice architecture inside the state and was imitated around the world. Recommender systems and targeted advertising perform population-scale behavioural prediction, generating what Zuboff calls prediction products. Earlier experiments anticipated the Prime Radiant more literally. Chile's Project Cybersyn (1971–73), designed with the cybernetician Stafford Beer, built an operations room to steer a national economy by real-time feedback (Eden Medina, *Cybernetic Revolutionaries*, 2011), and the Soviet OGAS project envisaged a nationwide computer network for economic planning that was never built (Benjamin Peters, *How Not to Network a Nation*, 2016). In each case the four marks of stochastic sovereignty recur: the object is a distribution of behaviour; the instrument is the modulation of information, defaults, and affect; the legitimacy is aggregate welfare or efficiency; and the condition, more or less avowed, is opacity, what Frank Pasquale called *The Black Box Society* (2015). The Lucas critique and Goodhart's law haunt every such system, since models deployed on the populations they describe change those populations.

The countermovement is a politics of publicity. The European Union's General Data Protection Regulation (adopted 2016, applicable from 2018) restricted decisions based solely on automated processing, and its Artificial Intelligence Act (adopted 2024) imposed transparency

and documentation duties on high-risk systems. Whatever their efficacy, these instruments share a principle: the existence, purpose, and logic of predictive governance must be disclosed to those it governs. They are attempts to impose Kant's test on stochastic sovereigns.

11.5 The Psychohistory of the Sentence

Large language models realise psychohistory's first axiom in a new medium. Trained on corpora of human writing at the scale of a civilisation, they predict the distribution of the next unit of text: reliable for the aggregate of usage, uncertain for any singular utterance, exactly as Seldon's science is reliable for quadrillions and silent about individuals. Their analogue of the second axiom is that the corpora on which they were first trained were produced by a population unaware that it was training a predictor. That condition is now being breached, as text is increasingly written with, for, or by such systems. Ilia Shumailov and colleagues ("AI Models Collapse When Trained on Recursively Generated Data," *Nature*, 2024) showed that models trained recursively on model-generated data progressively lose the tails of the original distribution and degrade, a phenomenon they named model collapse. In the Trilogy's vocabulary, once the population is aware of, and shaped by, the predictor, its behaviour ceases to be independent of the predictor, and the variance that made it informative contracts toward the predictor's own mean.

The first casualties of that contraction are the tails: the rare, the singular, the improbable, the material from which novelty arises. The Mule, in this light, is the figure of the tail, and the Trilogy's lesson for an information ecology saturated by prediction is that the singular must be protected not as noise but as the reservoir of the future. Model collapse is the Landauer cost of a self-referential prediction regime, paid in erased tails. The question of the Second Foundation, who governs the governor, becomes in this domain the question of who curates the corpus: who decides which data, which filters, and which norms shape a predictor that in turn shapes the population. Practices of disclosure such as model cards (Mitchell et al., 2019) and datasheets for datasets (Gebru et al., 2018) are, once again, applications of the publicity principle to a stochastic medium.

11.6 Kant's Publicity Test

The second appendix of Kant's *Perpetual Peace* states the transcendental formula of public right: actions affecting the rights of others are unjust if their maxim cannot be made public, and a maxim that must be kept secret to succeed, which I could not avow without provoking general resistance to my purpose, is unjust for that very reason. The Second Foundation fails this test by its own account. Its maxim, to govern the Galaxy covertly for the Galaxy's good, is effective only if secret, and the Trilogy's explicit rationale for secrecy is that disclosure provokes

resistance, as the revolt of the Darell circle demonstrates. Kant's test is therefore not an external standard imposed on the Trilogy but an exact description of the Second Foundation's self-understanding, converted into a verdict. John Rawls (*A Theory of Justice*, 1971) included publicity among the formal constraints of the concept of right: principles of justice must be capable of being publicly known and acknowledged. From Kant to Rawls, the liberal tradition makes publicity a condition of legitimacy, and by that standard the politics of the Second Foundation is constitutively illegitimate. That is why it makes so fascinating a fiction and so dangerous a charter.

Defenders of expert paternalism reply that some public goods require confidentiality: central banks deliberate privately, security services act covertly, and authorities withhold information to prevent panics, bank runs being Merton's own example of the self-fulfilling prophecy. The objection clarifies the test rather than defeating it. Kant's formula concerns maxims, not every operational detail. An independent central bank operating under "constrained discretion," the formula associated with Ben Bernanke, publishes its mandate, its targets, and in due course its deliberations; its existence and its agents are public even when its discussions are not. Such institutions are the Kydland–Prescott compromise between rule and discretion, conducted under publicity. The Second Foundation fails not because its deliberations are confidential but because its existence, its mandate, and its agents are secret. The line between legitimate delegated expertise and stochastic sovereignty runs exactly there.

CHAPTER

XII

Conclusion

Covenantal Contingency

> *"At the end of every seven years, in the solemnity of the year of release, in the feast of tabernacles... thou shalt read this law before all Israel in their hearing."*
>
> Deuteronomy 31:10–11, Authorized Version

12.1 The Argument Recapitulated

Augustine composed *The City of God* after the Visigothic sack of Rome in 410, to show that the fall of the earthly city could not touch the pilgrim city whose destiny lay elsewhere. The Foundation Trilogy, composed in a decade of world war, rewrites the two cities as two Foundations and makes the heavenly city a bureaucracy. This dissertation has traced how. The Trilogy begins under Laplace's demon, with a Plan that executes itself and a Founder present only as a recording; passes under Maxwell's, as the Second Foundation sorts historical trajectories at the price of information; and ends under Descartes's, as an order manufactures the beliefs of the governed about their own governance. The synthesis, the Seldonian Demon, is the agent of stochastic sovereignty.

Along the way, psychohistory was shown to return statistical reason to its theological birthplace in Arbuthnot and Süssmilch, to spatialise Comte's law of three stages, to literalise Smith's invisible hand by giving it fingers, and to settle the socialist calculation debate by fiat of scale. Its axiom of ignorance proved to be an arcanum concealing a choice of discretion over rule, since the fixed-point theorems of 1954 establish that self-confirming public predictions exist. The Mule emerged as the amplified singular, a quantum-genetic event at a Maxwellian singular point, Schmitt's exception, Dostoevsky's spite, and Hegel's unloved master, defeated by a single free act of recognition. The Second Foundation emerged as a providential machine, an angelology of Speakers, Joachim's order of spiritual men, and the Mule distributed, made invisible, and made fertile. Every reduction of historical entropy in the Trilogy carries a Landauer cost that the narrative never tallies. The comparative taxonomy showed the Trilogy passing

through all four modes of response to its own premise, and the analysis of Seldonism showed its charter in operation.

12.2 The Plan as Promise

What actually sustains the Plan inside the fiction? Not its mathematics, which the reader never sees and which fails outright before the Mule. What sustains it is fidelity. Hardin waits because he trusts that Seldon has foreseen; Mallow acts because he believes the Plan permits; the Foundation endures three centuries of crises because the Vault periodically reaffirms that its trials have a meaning. When the Vault speaks irrelevantly on the eve of the Mule's conquest, the Foundation collapses not because a computation has failed but because a promise appears to have been broken. The Second Foundation's closing stratagem restores not a prediction but a faith.

Psychohistory is a promise that has forgotten it is a promise. Among Austin's paradigm cases of the performative utterance is "I promise," the utterance that creates the obligation it names. The performativity of the Plan analysed in Chapter VIII is the performativity of promising: the Plan produces the future it announces by binding those who accept it. Its truth is not correspondence but fidelity, and its failure mode is not error but betrayal.

12.3 The Vault as Covenant Renewal

Read in this light, the apparitions in the Time Vault resemble less an oracle than a covenant renewal. Deuteronomy commands that at the end of every seven years, at the feast of Tabernacles in the year of release, the law be read aloud to the whole assembled people, "men, and women, and children, and thy stranger that is within thy gates" (31:12); the twenty-fourth chapter of Joshua gathers the tribes at Shechem to renew the covenant by public assent. A covenant binds the future through mutual promise across generations; its authority is fidelity rather than predictive accuracy; and it is renewed in public, in the hearing of all. The Time Vault inverts every term. Its promise is unilateral, its audience restricted to officials and notables, its content a forecast rather than a law, and its author beyond renegotiation. The Trilogy preserves the form of covenant while emptying it of reciprocity. Asimov's secular Jewish formation need not have supplied the model for the homology to instruct: the Seldon Plan is a covenant with only one party. It is also, as Chapter III argued, Leviathan without covenant: where Hobbes's sovereign is authorised by the mutual promise of those it governs, Seldon's is authorised by no promise but its own.

The Deuteronomic text contains a further detail of decisive importance. The year of release in which the law is read is the sabbatical year in which debts are remitted (Deuteronomy 15:1–

2). The public renewal of the promise and the release of obligations are fixed to the same moment. The ancient calendar joins promising and forgiving in a single institution.

12.4 Promising and Forgiving

Hannah Arendt's *The Human Condition* (1958) argues that action, because it takes place among plural agents each capable of beginning something new, is boundless, unpredictable, and irreversible. She identifies two faculties as remedies internal to action itself. Forgiving answers irreversibility, releasing agents from the consequences of what they have done. Promising answers unpredictability, binding agents to a future none can foresee and so establishing "islands of predictability" in an ocean of uncertainty. Arendt sets promising against the alternative remedy for unpredictability, mastery through the domination of oneself and others, and in "What Is Freedom?" draws the conclusion: "If men wish to be free, it is precisely sovereignty they must renounce."

Read through Arendt, the Trilogy stages a choice between the two remedies for unpredictability, the sovereignty of the Second Foundation, which masters plurality by mastering minds, and the promise, which the Plan contains in suppressed form. Arendt's concept of natality, the capacity to begin that is given with every birth, illuminates the Mule and Bayta as the dark and light faces of beginning. The Mule is natality without plurality, a beginning that abolishes the capacity of others to begin. Bayta is natality within plurality, a beginning that preserves the free recognition on which it depends. Psychohistory is the attempt to master natality statistically, and the Second Foundation is what that attempt becomes when it succeeds.

12.5 Covenantal Contingency

Against stochastic sovereignty this dissertation proposes *covenantal contingency*: a polity's capacity to remain open to the singular while sustaining continuity across generations through publicly avowed, revisable promises rather than concealed prediction. Five principles define it.

The first is *publicity*. The existence, mandate, and agents of any intergenerational steering must be public, even where particular deliberations remain confidential, in accordance with Kant's transcendental formula and Rawls's publicity condition.

The second is *revisability*. The promise must be amendable by those it binds: a covenant renewed by assent, not a Plan executed by fidelity alone. Popper's piecemeal method belongs here, but conducted in the open, where its errors can be seen and corrected by those who bear them.

The third is *natality*. Institutions must protect the capacity for unprecedented action, the tails of the distribution. The possibility of a Mule is the price of the possibility of a Bayta, and a civilisation that eliminates the first will eliminate the second.

The fourth is *avowed cost*. The costs of steering must be declared and, so far as possible, borne by those who choose them. There can be no hidden erasures; the Landauer ledger must be kept in public.

The fifth is *forgiveness*. Because action is irreversible and every steering will err, a covenantal polity requires institutions of release: amnesty, reconciliation, the remission of debts. Here the Trilogy offers its most exact self-diagnosis, for it contains no forgiveness at all. The Mule is not forgiven but rewritten; the fifty are not mourned but spent; the deceived are not reconciled but kept ignorant. Stochastic sovereignty cannot forgive, because forgiveness is addressed to a *who*, and psychohistory knows only a *how many*.

12.6 Limits and Further Work

The study has treated the Trilogy as published between 1951 and 1953 and the later sequels as apocrypha. A full account of Asimov's late synthesis of robots and Foundation, in which R. Daneel Olivaw and the collective mind of Gaia displace the Plan, would test whether the turn to collective consciousness resolves stochastic sovereignty or merely relocates it. The gendering of the Trilogy's hinge points also deserves sustained treatment. In a work almost devoid of women in its first volume, the decisive interventions at its turning points belong to women: Bayta against the Mule, Arcadia as the unwitting courier of the Second Foundation's misdirection, Lady Callia as its hidden agent on Kalgan. This is a pattern the Trilogy's sexual politics cannot theorise and that the critiques of Joanna Russ (1970) and Ursula K. Le Guin open to analysis. The comparative reception of psychohistory in Soviet and Chinese science fiction, from the Strugatskys to Liu Cixin, merits a monograph of its own, and the empirical study of Seldonism among economists, technologists, and philanthropists remains to be undertaken.

12.7 Coda: The Empty Vault

Benjamin's automaton wins every game because a hidden theologian guides its hand. The Trilogy shows what happens when the dwarf takes over the machine: the puppet goes on winning, and no one any longer knows who is playing. Read against itself, the Trilogy suggests that the task is not to expel the dwarf, to abandon the theological dimension of long-term commitment and of faith across generations, but to bring him out from beneath the table, so that theology appears not as concealed calculation but as public covenant.

The image with which this dissertation closes is not in the Trilogy, though the Trilogy implies it: a Time Vault whose glass cubicle stands empty, because the Foundation no longer needs the apparition of its Founder, having learned to make, renew, and when necessary release its own promises. The end of the Plan would be the beginning of politics. A civilisation is not saved by being predicted. If it is saved at all, it is saved by being promised.

APPARATUS

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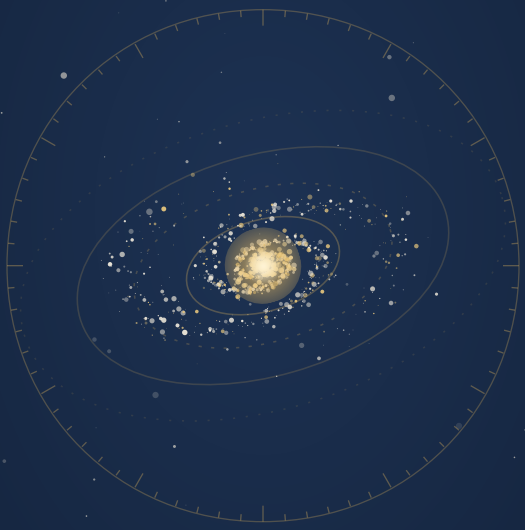
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APPARATUS

Keywords

Foundation Trilogy ♦ Isaac Asimov ♦ psychohistory ♦ stochastic sovereignty ♦ Seldonian Demon ♦ philosophy of history ♦ political theology ♦ providential machine ♦ secularization ♦ reoccupation ♦ Joachimism ♦ eschatology ♦ theodicy ♦ arcana imperii ♦ esotericism ♦ reflexive prediction ♦ self-fulfilling prophecy ♦ Oedipus effect ♦ fixed-point theorem ♦ rules versus discretion ♦ performativity ♦ governmentality ♦ instrumentarian power ♦ cybernetics ♦ law of requisite variety ♦ Maxwell's demon ♦ Landauer's principle ♦ entropy ♦ statistical mechanics ♦ Laplacean determinism ♦ sovereign exception ♦ great man theory ♦ Hegelian recognition ♦ ressentiment ♦ doux commerce ♦ Saint-Simonianism ♦ positivism ♦ socialist calculation debate ♦ Gibbon ♦ Montesquieu ♦ Ibn Khaldun ♦ Vico ♦ Spengler ♦ Toynbee ♦ longue durée ♦ cliodynamics ♦ longtermism ♦ model collapse ♦ cryptotopia ♦ mnemopolitics ♦ covenant ♦ natality ♦ covenantal contingency ♦ Arendt ♦ historical inevitability ♦ Leviathan ♦ mathematical sublime ♦ end of history ♦ bureaucratic rationality ♦ Holocaust historiography ♦ science fiction studies



A civilisation is not saved by being predicted.
If it is saved at all, it is saved by being promised.

THE AUTOMATON AND THE DWARF